
Robust Constrained Offline Reinforcement Learning with Linear Function Approximation

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Abstract

Due to sim-to-real gaps, the goal of reinforcement learning is not only to maximize cumulative rewards but also to ensure that the learned policy can be safely deployed and remains robust to potential perturbations in the real environment. To this end, this paper considers *robust constrained linear Markov decision problem (Lin-RCMDPs)*, where an agent aims to optimize the expected total rewards and ensure constraint satisfaction against the worst-case performance within an unknown uncertainty set characterized by the total variation distance using batch data. We develop a sample-efficient model-based algorithm with a primal-dual approach. Under minimal data coverage assumptions, we establish a sample complexity bound and extend the analysis to model misspecification by relaxing the linear MDP assumption. To the best of our knowledge, this provides the first provably near-optimal offline RL sample complexity bound for Lin-RCMDPs.

1 Introduction

In reinforcement learning (RL), agents aim to learn an optimal policy through interactions with the environment, continuously refining their strategy to maximize the expected cumulative reward [1]. However, many practical RL applications, such as autonomous driving [2–5], healthcare [6–8], wireless security [9–11], and robotics [12–14], are subject to additional constraints related to limited system resources or safety concerns. For example, autonomous robots must avoid collision [15, 16], medical applications often face cost limitations [17, 18], and financial management is subject to legal and business restrictions [19]. To address such challenges, Constrained Markov Decision Processes (CMDPs) have attracted increasing interest, which aim to maximize expected utility while ensuring that the resulting policy satisfies specific constraints [20–26].

For solving CMDPs, existing approaches [27–29] find feasible solutions by interacting with the environment in an online manner, while such data collection may be both costly and potentially hazardous in many real-world applications. To address this issue, this paper focuses on learning policies from historical (or batch) data, where significant advancements have been made in both theoretical and practical aspects of offline RL without constraints [30–36]. However, a key assumption remains that the test environment is identical to the training environment. This assumption is often impractical in real-world settings, where even minor environmental changes can cause significant performance degradation. Consequently, developing robust policies that can handle such environmental uncertainties has become an urgent research problem.

In response, distributionally robust constrained RL methods have garnered significant attention [21, 25, 37], which optimize the worst-case performance of the policy based on historical data from the nominal environment. However, existing approaches are confined mainly to the online setting [21, 25, 37] or focus primarily on tabular models with finite state and action spaces [25]. In these

	State Representation	Blanchet et al. [36]	Wang et al. [40]	Ghosh [25]	This Work
unconstraint	$S \times A$ -rectangular (Tabular)	✓	✓	✓	✓
	d -rectangular (Linear)	✓	✓	✗	✓
constraint	$S \times A$ -rectangular (Tabular)	✗	✗	✓!	✓
	d -rectangular (Linear)	✗	✗	✗	✓

Table 1: Comparison with the most relevant works in robust RL. ✓ indicates that the work is capable of addressing the model with robust partial coverage data, ✓! signifies that the work requires full coverage data to solve the model, and ✗ denotes that the work is not applicable to the model. Light green highlights the models that are either introduced or proven to be tractable in this work.

settings, the sample complexity scales linearly with the size of the state-action space, which becomes prohibitively expensive for high-dimensional problems. Thus, a natural question that arises is:

Is it possible to develop a provably sample-efficient offline RL algorithm with linear function approximation that ensures both distributional robustness and safety?

1.1 Contribution

In this work, we give an affirmative answer to the open problem of learning robust constrained linear MDPs from purely offline data. Specifically, we focus on a family of finite-horizon distributionally robust constrained linear MDPs (Lin-RCMDPs), where the uncertainty sets are motivated both by practical considerations [38] and by recent theoretical works [39, 40]. Our main contributions are summarized as follows:

- We develop a distributionally robust variant of constraint optimistic-pessimistic least-squares value iteration, referred to as CRQP-VI. By integrating linear MDP model representations with a data-driven penalty term, CRQP-VI effectively mitigates the scarcity of offline data.
- We provide the first finite-sample guarantees for Lin-RCMDPs under minimal data assumptions. Moreover, under the partial feature coverage assumption of offline data, CRQP-VI learns an ϵ -optimal policy and ensures constraint violations at most ϵ , as long as the sample size satisfies $\tilde{O}(C_{\text{rob}}^* d^2 H^4 / \epsilon^2)$ (cf. Corollary 1), where C_{rob}^* is a clipped concentrability coefficient characterizing partial feature coverage of the offline data. In addition, under full feature coverage, the sample complexity improves to $\tilde{O}(dH^4 / \kappa^{-1} \epsilon^{-2})$ (cf. Corollary 2), where κ quantifies the degree of full feature coverage. Significantly, this matches the complexity bounds known for the unconstrained robust counterpart [40].
- We further extend our theoretical framework to encompass model misspecification in the transition model—a realistic scenario from inaccuracies in the soft-state aggregation model. Our results confirm that CRQP-VI maintains robust performance guarantees even when the MDP transition dynamics deviate from the idealized linear assumptions.

A comprehensive comparison with relevant works is presented in Table 1.

1.2 Related works

Constrained standard and linear MDPs. Primal-dual algorithms are widely used to CMDPs, with tight performance guarantees established [41, 21, 22, 37]. These methods leverage strong duality results to demonstrate convergence to the optimal solutions. Early analyses in the tabular settings delivered rigorous convergence and sample-complexity bounds [42, 20, 43–47], and subsequent efforts extended these results to more structured models such as linear-kernel MDPs [48], whose distinctions from linear MDPs are expounded in [21]. While recent studies have targeted linear MDP [49, 37, 21], they neglected robustness: policies that perform well in expectation can violate constraints when dynamics are misspecified. Robust CMDPs are fundamentally harder because the worst-case transition kernel depends on the policy, rendering the feasible occupancy-measure set non-convex and thwarting standard analyses.

Distributionally robust RL with linear MDPs. Robust linear MDPs (Lin-RMDPs) have recently attracted considerable attention [36, 50, 40]. Early progress focused on offline learning with total variation (TV) uncertainty balls, where [36] provided the first sample complexity guarantees and

[40] sharpened the analysis for the same unconstrained TV setting studied here. The result in [40] showed a tighter sample complexity dependency on the feature dimension d , matching standard offline linear MDPs [33, 51]. Beyond TV distance considered herein, distributional robustness has been considered under Kullback-Leibler (KL) divergence [50, 36]. In the online setting, [52] showed that near-optimal regret can be achieved under additional structure on the uncertainty set, while [53] restricted attention to offline learning with the well-explored data coverage assumption. Complementary to linear function approximation, [54, 55] have investigated richer model classes.

Distributionally robust RL with constraints. Early work on RCMDPs has relied on a primal-dual framework [56, 24, 23]. [56, 24] proposed heuristic schemes that alternately compute the worst-case return over a prescribed uncertainty set and improve the policy with standard techniques such as policy gradients. [23] introduced a robust primal-dual algorithm and showed that strong duality can fail even when a strictly feasible policy exists (aka Slater’s condition), precluding sample-complexity guarantees for these heuristics. More recently, [26] employed a trust-region update that maintains performance and feasibility across iterations, though convergence to a near-optimal policy is still unproven. In the tabular setting, [25] established the first finite-sample bounds, while [57] developed a policy-gradient method based on an epigraph form of the RCMDP problem.

Notation. For any finite set $\mathcal{S}$, let $\Delta(\mathcal{S})$ be its probability simplex. Given integers $H, d \geq 1$, write $[H] := \{1, \dots, H\}$ and $[d] := \{1, \dots, d\}$. Denote $\mathbb{1}_i$ as the vector of appropriate dimension, where the i -th entry is 1 and all other entries are 0. The identity matrix of size d is represented as I_d . For any vectors $x, y \in \mathbb{R}^d$, let $\|x\|_2$ and $\|x\|_1$ denote the l_2 and l_1 norm of x , and $\langle x, y \rangle$ represent their inner product; for any positive semi-definite matrix $A \in \mathbb{R}^{d \times d}$, define $\|x\|_A \equiv \sqrt{x^\top A x}$. The cardinality of a set $\mathcal{D}$ is $|\mathcal{D}|$. Furthermore, we use $\min\{a, b\}_+$ to denote the minimum of a and b when $a, b > 0$, and 0 otherwise. The standard notation $\mathcal{O}(\cdot)$ is adopted to represent the orderwise scaling of a function. The variant $\tilde{\mathcal{O}}(\cdot)$ is identical except that it suppresses logarithmic factors.

2 Problem Setup

In this section, we introduce the formulation of *distributionally robust constrained Markov Decision Processes (RCMDPs)*, along with the corresponding learning objective and assumptions regarding the model and batch data. To begin with, we consider standard constrained MDPs (CMDPs) as follows.

CMDPs: standard constrained MDPs. Consider a standard constrained MDP, denoted by $\mathcal{M} = (\mathcal{S}, \mathcal{A}, H, P = \{P_h\}_{h=1}^H, r = \{r_h\}_{h=1}^H, g = \{g_h\}_{h=1}^H)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, and H is the horizon length. At each step $h \in [H]$, $P_h : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ is the transition kernel, $r_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is the deterministic reward function and $g_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is the utility function. In addition, we denote $\pi = \{\pi_h\}_{h=1}^H$ as the policy of the agent, where $\pi_h : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ is the action selection probability over the action space $\mathcal{A}$ at the step h . Given any policy π in the CMDP, the learning objective is to solve the following optimization problem:

$$\max_{\pi} \mathbb{E}_{s_0 \sim \zeta} \left[V_{r,1}^{\pi,P}(s_0) \right] \quad \text{subjection to } \mathbb{E}_{s_0 \sim \zeta} \left[V_{g,1}^{\pi,P}(s_0) \right] \geq b,$$

where b is some positive threshold, ζ is initial state distribution, and for any state $s \in \mathcal{S}$ and $h \in [H]$, $V_{r,h}^{\pi,P}(s) = \mathbb{E}_{\pi,P} \left[\sum_{t=h}^H r_t(s_t, a_t) \mid s_h = s \right]$ and $V_{g,h}^{\pi,P}(s) = \mathbb{E}_{\pi,P} \left[\sum_{t=h}^H g_t(s_t, a_t) \mid s_h = s \right]$ are value functions for the reward function r and utility function g , respectively.

Linear function approximation. To handle the potentially massive state space, we impose the following linear model assumption, which is commonly used in the previous literature [49, 21, 37, 40].

Assumption 1 (Linear CMDPs). *A finite-horizon CMDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, H, P, r, g)$ is a linear CMDP if given a known feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$, there exist d unknown measures $\mu_h^P = (\mu_{h,1}^P, \dots, \mu_{h,d}^P)$ over the state space $\mathcal{S}$ and two unknown vectors $\theta_{r,h}, \theta_{g,h} \in \mathbb{R}^d$ at each step h such that for $\forall (h, s, a, s') \in [H] \times \mathcal{S} \times \mathcal{A} \times \mathcal{S}$,*

$$r_h(s, a) = \langle \phi(s, a), \theta_{r,h} \rangle, \quad g_h(s, a) = \langle \phi(s, a), \theta_{g,h} \rangle, \quad P_h(s' \mid s, a) = \langle \phi(s, a), \mu_h^P(s') \rangle.$$

Without loss of generality, we assume that $\|\phi(s, a)\|_2 \leq 1$ and $\phi_i(s, a) \geq 0$ for any $(s, a, i) \in \mathcal{S} \times \mathcal{A} \times [d]$, and $\max \left\{ \int_{\mathcal{S}} \|\mu_h^P(s)\|_2 ds, \|\theta_{r,h}\|_2, \|\theta_{g,h}\|_2 \right\} \leq \sqrt{d}$ for all $h \in [H]$.

Lin-RCMDPs: distributionally robust linear CMDPs. To handle the uncertainty of environments, we consider *distributionally robust* linear CMDPs (Lin-RCMDPs), where the transition kernel can be an arbitrary one within an uncertainty set around the nominal kernel [21]. Formally, we denote it by $\mathcal{M}_{\text{rob}} = (\mathcal{S}, \mathcal{A}, H, \mathcal{P}^\rho(P^0), r, g)$, where P^0 represents a nominal transition kernel and then $\mathcal{P}^\rho(P^0)$ represents the uncertainty set (a ball) around the nominal P^0 with some uncertainty level $\rho \geq 0$. We assume that the uncertainty set satisfies the following d -rectangularity assumption [58, 40].

Assumption 2 (d -rectangularity of the uncertainty set). *In Lin-RCMDPs, the uncertainty set $\mathcal{P}^\rho(P^0)$ is d -rectangular, i.e., $\mu_{h,i}^0 \in \Delta(\mathcal{S})$ for any $(h, i) \in [H] \times [d]$ and*

$$\mathcal{P}^\rho(P^0) := \otimes_{[H], \mathcal{S}, \mathcal{A}} \mathcal{P}^\rho(P_{h,s,a}^0), \quad \text{with } \mathcal{P}^\rho(P_{h,s,a}^0) := \{\langle \phi(s, a), \mu_h(\cdot) \rangle : \mu_h \in \mathcal{U}^\rho(\mu_h^0)\},$$

where $\mu_h^0 := \mu_h^{P^0}$ for simplicity, $\mathcal{U}^\rho(\mu_h^0) := \otimes_{[d]} \mathcal{U}^\rho(\mu_{h,i}^0)$, $\mathcal{U}^\rho(\mu_{h,i}^0) := \{\mu_{h,i} : D_{\text{TV}}(\mu, \mu_{h,i}^0) \leq \rho \text{ and } \mu_{h,i}^0 \in \Delta(\mathcal{S})\}$, $\otimes_{[d]}$ (resp. $\otimes_{[H], \mathcal{S}, \mathcal{A}}$) denotes Cartesian products over $[d]$ (resp. $[H], \mathcal{S}$, and $\mathcal{A}$) and ρ is the uncertainty level. Here, $D_{\text{TV}}(\mu_{h,i}, \mu_{h,i}^0) = \frac{1}{2} \|\mu_{h,i} - \mu_{h,i}^0\|_1$ represent the total variation (TV) distance between two probability measures.

Note that when the uncertainty radius $\rho = 0$, Lin-RCMDPs reduce to the standard linear CMDPs satisfying Assumption 1. To further robustify the learned policy when $\rho > 0$, we are interested in the *worst-case* performance induced by all possible transition kernels over the uncertainty set, characterized by *robust value function* (resp. *robust Q function*) for the reward r and utility g , respectively. For simplicity, we interchangeably denote that for $j = r, g$,

$$\forall (h, s, a) \in [H] \times \mathcal{S} \times \mathcal{A} : V_{j,h}^{\pi, \rho}(s) := \inf_{P \in \mathcal{P}^\rho(P^0)} V_{j,h}^{\pi, P}(s), \quad Q_{j,h}^{\pi, \rho}(s, a) = \inf_{P \in \mathcal{P}^\rho(P^0)} Q_{j,h}^{\pi, P}(s, a).$$

Additionally, we denote $\pi^* = \{\pi_h^*\}_{h=1}^H$ as a deterministic *optimal robust policy* [59] that maximizes the robust value function for reward r and satisfies the constraint simultaneously, i.e., $\pi_h^*(s) = \arg \max V_{r,h}^{\pi, \rho}(s)$ and $V_{g,h}^{\pi, \rho}(s) \geq b$ for any $(s, h) \in \mathcal{S} \times [H]$. The resulting *optimal robust value/utility function* and *optimal robust Q-function* are defined as:

$$V_{r,h}^{*, \rho}(s) := V_{r,h}^{\pi^*, \rho}(s) = \max_{\pi} V_{r,h}^{\pi, \rho}(s), \quad \forall \in [H] \times \mathcal{S},$$

$$V_{g,h}^{*, \rho}(s) := V_{g,h}^{\pi^*, \rho}(s) = \inf_{P \in \mathcal{P}^\rho(P^0)} \mathbb{E}_{\pi^*, P} \left[\sum_{t=h}^H g_t(s_t, a_t) \mid s_h = s \right], \quad \forall \in [H] \times \mathcal{S},$$

and for any $(h, s, a) \in [H] \times \mathcal{S} \times \mathcal{A}$,

$$Q_{r,h}^{*, \rho}(s, a) := Q_{r,h}^{\pi^*, \rho}(s, a) = \max_{\pi} Q_{r,h}^{\pi, \rho}(s, a), \quad Q_{g,h}^{*, \rho}(s, a) := Q_{g,h}^{\pi^*, \rho}(s, a) = Q_{g,h}^{\pi^*, \rho}(s, \pi^*(s)).$$

In addition, conditioned on some initial state distribution ζ , we define the induced *occupancy distribution* w.r.t. any policy π and transition kernel P as follows: for any $(h, s, a) \in [H] \times \mathcal{S} \times \mathcal{A}$,

$$d_h^{\pi, P}(s) := d_h^{\pi, P}(s; \zeta) = \mathbb{P}(s_h = s \mid \zeta, \pi, P), \quad d_h^{*, P}(s, a) = d_h^{\pi^*, P}(s) \pi_h^*(a \mid s). \quad (1)$$

Similar to (1), we also denote the occupancy distribution associated with the optimal robust policy π^* and some transition kernel P by

$$\forall (h, s, a) \in [H] \times \mathcal{S} \times \mathcal{A} : d_h^{*, P}(s) := d_h^{\pi^*, P}(s; \zeta), \quad d_h^{*, P}(s, a) := d_h^{\pi^*, P}(s) \pi_h^*(a \mid s).$$

Robust linear Bellman operator and optimality equations. A key component of Lin-RCMDPs is the extension of the Bellman optimality principle for both reward and utility functions, captured by the following *robust Bellman consistency equation* (or equivalently, the *robust Bellman optimality equation* for π^*): for $j = r, g$,

$$\forall (h, s, a) \in [H] \times \mathcal{S} \times \mathcal{A} : Q_{j,h}^{\pi, \rho}(s, a) = \mathbb{B}_{j,h}^\rho V_{j,h+1}^{\pi, \rho}(s, a), \quad V_{j,h}^{\pi, \rho}(s) = \mathbb{E}_{a \sim \pi_h(\cdot \mid s)} [Q_{j,h}^{\pi, \rho}(s, a)].$$

where the robust Bellman operator and transition operator for any function $f : \mathcal{S} \rightarrow \mathbb{R}$ are defined by

$$[\mathbb{B}_{j,h}^\rho f](s, a) := j_h(s, a) + [\mathbb{P}_h^\rho f](s, a), \quad (2)$$

$$[\mathbb{P}_h^\rho f](s, a) := \inf_{\mu_h \in \mathcal{U}^\rho(\mu_h^0)} \int_{\mathcal{S}} \langle \phi(s, a), \mu_h(s') \rangle f(s') ds'.$$

Note that under Assumption 2, the robust Bellman operator inherits the linearity of the Bellman operator in standard linear MDPs [42, Proposition 2.3], as shown in the following lemma.

Lemma 1 (Linearity of robust Bellman operators). *[Lemma 1 in [40]] Suppose that the finite-horizon Lin-RCMDPs satisfies Assumption 1 and 2. There exist weights $w_j^\rho = \{w_{j,h}^\rho\}_{h=1}^H$, where $w_{j,h}^\rho := \theta_{j,h} + \inf_{\mu_h \in \mathcal{U}^\rho(\mu_h^0)} \int_{\mathcal{S}} \mu_h(s') f(s') ds'$ for any $h \in [H]$, such that $\mathbb{B}_{j,h}^\rho f(s, a)$ is linear with respect to the feature map ϕ , i.e., $\mathbb{B}_{j,h}^\rho f(s, a) = \langle \phi(s, a), w_{j,h}^\rho \rangle$ for $j = r, g$.*

Offline data. Consider a batch dataset $\mathcal{D} = \{(s_h^\tau, a_h^\tau, r_h^\tau, g_h^\tau, s_{h+1}^\tau)\}_{h \in [H], \tau \in [K]}$ consisting of K i.i.d. trajectories generated by executing some (mixed) behavior policy $\pi^b = \{\pi_h^b\}_{h=1}^H$ in interaction with a nominal linear MDP $\mathcal{M}^0 = (\mathcal{S}, \mathcal{A}, H, P^0, r, g)$. Note that $\mathcal{D}$ contains KH transition-reward sample tuples, where each sample tuple $(s_h^\tau, a_h^\tau, r_h^\tau, g_h^\tau, s_{h+1}^\tau)$ represents that the agent took the action a_h^τ in state s_h^τ , received the reward $r_h^\tau = r_h(s_h^\tau, a_h^\tau)$ and the utility $g_h^\tau = g_h(s_h^\tau, a_h^\tau)$, and then transitioned to the next state $s_{h+1}^\tau \sim P_h^0(\cdot | s_h = s_h^\tau, a_h = a_h^\tau)$. We further define the slice of data at step h as

$$\mathcal{D}_h^0 = \{(s_h^\tau, a_h^\tau, r_h^\tau, g_h^\tau, s_{h+1}^\tau)\}_{\tau \in [K]}.$$

For simplicity, we abuse the notation $\tau \in \mathcal{D}_h^0$ to denote $(s_h^\tau, a_h^\tau, r_h^\tau, g_h^\tau, s_{h+1}^\tau) \in \mathcal{D}_h^0$. In addition, we define the occupancy distribution induced by the behavior policy π^b and the nominal transition kernel P^0 at each step h , conditioned on the initial state distribution ζ , as:

$$d_h^b(s) := d_h^{\pi^b, P^0}(s; \zeta) \quad \text{and} \quad d_h^b(s, a) := d_h^{\pi^b, P^0}(s, a; \zeta), \quad \forall (h, s, a) \in [H] \times \mathcal{S} \times \mathcal{A}.$$

Learning goal. Given the batch dataset $\mathcal{D}$ and an initial state distribution ζ , the Lin-RCMDP $\mathcal{M}_{\text{rob}}$ seeks to solve the following problem:

$$\max_{\pi} \mathbb{E}_{s_0 \sim \zeta} [V_{r,1}^{\pi, \rho}(s_0)] \quad \text{subjection to } \mathbb{E}_{s_0 \sim \zeta} [V_{g,1}^{\pi, \rho}(s_0)] \geq b. \quad (3)$$

Our goal is to learn an ϵ -optimal and ϵ -level constrained robust policy $\hat{\pi}$ satisfies

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) := V_{r,1}^{\star, \rho}(\zeta) - V_{r,1}^{\hat{\pi}, \rho}(\zeta) \leq \epsilon, \quad \text{Violation}(\hat{\pi}; \zeta, \mathcal{P}^\rho) := b - V_{g,1}^{\hat{\pi}, \rho}(\zeta) \leq \epsilon, \quad (4)$$

with as few samples as possible, where ϵ represents the targeted accuracy, and

$$V_{r,1}^{\star, \rho}(\zeta) = \mathbb{E}_{s_1 \sim \zeta} [V_{r,1}^{\star, \rho}(s_1)], \quad V_{j,1}^{\hat{\pi}, \rho}(\zeta) = \mathbb{E}_{s_1 \sim \zeta} [V_{j,1}^{\hat{\pi}, \rho}(s_1)], \quad j \in \{r, g\}.$$

3 Solution Methodology

In this section, we first consider a rectified form of problem (3) using the dual variable. Then, we introduce a novel model-based algorithm, called *Constrained Robust Optimistic-Pessimistic Value Iteration* (CROP-VI), which constructs an empirical Bellman operator for Lin-RCMDPs in an offline fashion.

3.1 Rectified dual form

As highlighted by [25], the traditional primal-dual method developed for non-robust CMDPs does not directly extend to the robust CMDPs, since the robust state-action occupancy measure is not convex and strong duality does not hold. A feasible solution is to transform the original problem (3) into the following rectified dual form:

$$\max_{\pi} V_{r,1}^{\pi, \rho}(\zeta) - \beta (b - V_{g,1}^{\pi, \rho}(\zeta))_+, \quad (5)$$

where $\beta > 0$ is the penalty coefficient and $(x)_+ = \max\{x, 0\}$. The equivalence between the rectified formulation (5) and the original problem (3) is established in the following lemma.

Lemma 2. *Given $\epsilon > 0$, setting $\beta = H/\epsilon$ ensures that the optimal solution $\hat{\pi}$ of (5) incurs a constraint violation of at most ϵ , i.e., $(b - V_{g,1}^{\hat{\pi}, \rho}(\zeta)) \leq \epsilon$. Consequently, with $\beta = H/\epsilon$, if for any infeasible policy π such that $V_{g,1}^{\pi, \rho}(\zeta) - b < \epsilon$, then $\pi^\star$ (i.e., the optimal solution of (3)) is also optimal for (5).*

Lemma 2 establishes that the equivalence between (3) and (5) holds if there exists an ϵ gap in the value function between feasible and infeasible policies. In particular, as $\beta \rightarrow \infty$, the optimal solution of (5) converges to that of the original problem (3).

Algorithm 1: CROP-VI: Constrained Robust Optimistic-Pessimistic Value Iteration

Input: Dataset $\mathcal{D}$, feature map $\phi(s, a)$ for all $(s, a) \in \mathcal{S} \times \mathcal{A}$, performance tolerance ϵ , parameters $\beta = H/\epsilon$, $\lambda_0, \gamma_0 > 0$;

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1 Construct a temporally independent dataset  $\mathcal{D}^0 = \text{Two-fold-subsampling}(\mathcal{D})$  (Algorithm 2).
2 Initialization:  $\hat{V}_{r,H+1}(\cdot) = 0, \hat{V}_{g,H+1}(\cdot) = 0$ 
3 for steps  $h = H, H-1, \dots, 1$  do
4    $\Lambda_h = \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \phi(s_h^\tau, a_h^\tau)^\top + \lambda_0 I_d$ ; // sample covariance matrix
5    $\hat{\theta}_{r,h} = \Lambda_h^{-1} \left( \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) r_h^\tau \right)$ ;
6    $\hat{\theta}_{g,h} = \Lambda_h^{-1} \left( \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) g_h^\tau \right)$ ;
7   for feature  $i = 1, \dots, d$  do
8     | Update  $\hat{v}_{h,i}^{\rho, \hat{V}_r}$  and  $\hat{v}_{h,i}^{\rho, \hat{V}_g}$  via (8).
9   end
10   $\hat{w}_{r,h}^{\rho, \hat{V}_r} = \hat{\theta}_{r,h} + \hat{v}_{h,i}^{\rho, \hat{V}_r}, \hat{w}_{g,h}^{\rho, \hat{V}_g} = \hat{\theta}_{g,h} + \hat{v}_{h,i}^{\rho, \hat{V}_g}$ ;
11   $\bar{Q}_{r,h}(\cdot, \cdot) = \phi(\cdot, \cdot)^\top \hat{w}_{r,h}^{\rho, \hat{V}_r} - \gamma_0 \sum_{i=1}^d \|\phi_i(\cdot, \cdot)\|_{\Lambda_h^{-1}}$ ; // Pessimism
12   $\hat{Q}_{r,h}(\cdot, \cdot) = \min \{ \bar{Q}_{r,h}(\cdot, \cdot), H - h + 1 \}$ ; // Truncation
13   $\bar{Q}_{g,h}(\cdot, \cdot) = \phi(\cdot, \cdot)^\top \hat{w}_{g,h}^{\rho, \hat{V}_g} + \gamma_0 \sum_{i=1}^d \|\phi_i(\cdot, \cdot)\|_{\Lambda_h^{-1}}$ ; // Optimism
14   $\hat{Q}_{g,h}(\cdot, \cdot) = \min \{ \bar{Q}_{g,h}(\cdot, \cdot), H - h + 1 \}$ ; // Truncation
15   $\hat{\pi}_h(\cdot) = \arg \max_{a \in \mathcal{A}} \hat{Q}_{r,h}(\cdot, a) - \beta(b - \hat{Q}_{g,h}(\cdot, a))_+$ ; // Optimization
16   $\hat{V}_{r,h}^{\hat{\pi}}(\cdot) = \hat{Q}_{r,h}(\cdot, \hat{\pi}(\cdot)), \hat{V}_{g,h}^{\hat{\pi}}(\cdot) = \hat{Q}_{g,h}(\cdot, \hat{\pi}(\cdot))$ ; // Evaluation
17 end
Output:  $\hat{V}_r := \{\hat{V}_{r,h}^{\hat{\pi}}\}_{h=1}^{H+1}, \hat{V}_g := \{\hat{V}_{g,h}^{\hat{\pi}}\}_{h=1}^{H+1}, \hat{\pi} := \{\hat{\pi}_h\}_{h=1}^H$ 
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3.2 Empirical robust Bellman operator and strong duality

Recalling the robust Bellman operator given in (2), it can be equivalently expressed in its dual form:

$$(\mathbb{B}_{j,h}^\rho V_j)(s, a) = \langle \phi(s, a), \theta_{j,h} + \nu_h^{\rho, V_j} \rangle,$$

from the view of strong duality [59, 39, 40]. Here, $\nu_h^{\rho, V_j} = [\nu_{h,1}^{\rho, V_j}, \nu_{h,2}^{\rho, V_j}, \dots, \nu_{h,d}^{\rho, V_j}]^\top \in \mathbb{R}^d$ and its i -th coordinate is defined by

$$\nu_{h,i}^{\rho, V_j} := \max_{\alpha \in [\min_s V_j(s), \max_s V_j(s)]} \left\{ \mathbb{E}_{s \sim \mu_h^0} [V_j]_\alpha(s) - \rho(\alpha - \min_{s'} [V_j]_\alpha(s')) \right\}, \quad (6)$$

with $[V_j]_\alpha(s) = \min\{V_j(s), \alpha\}$. Since we do not have direct access to the nominal linear MDP $\mathcal{M}^0$ (i.e., $\theta_{j,h}$ and μ_h^0), we cannot perform value iteration directly with the above Bellman operator. To address this, we estimate $\hat{\theta}_{j,h} \in \mathbb{R}^d$ and $\hat{v}_h^{\rho, V_j} \in \mathbb{R}^d$ using the batch dataset $\mathcal{D}_h^0$ containing all the samples at h -th step in $\mathcal{D}^0$. To ensure numerical stability, we use ridge regression instead of directly minimizing empirical mean squared errors. Specifically, for any value function $V_j : \mathcal{S} \rightarrow [0, H]$ and time step $h \in [H]$, the estimator $\hat{\theta}_{j,h}$ and the i -th coordinate of $\hat{v}_h^{\rho, V_j}$ are defined as:

$$\hat{\theta}_{j,h} = \arg \min_{\theta \in \mathbb{R}^d} \sum_{\tau \in \mathcal{D}_h^0} (\phi(s_h^\tau, a_h^\tau)^\top \theta_j - j_h^\tau)^2 + \lambda_0 \|\theta_j\|_2^2 = \Lambda_h^{-1} \left(\sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) j_h^\tau \right), \quad (7)$$

$$\hat{v}_{h,i}^{\rho, V_j} = \max_{\alpha \in [\min_s V_j(s), \max_s V_j(s)]} \left\{ \hat{v}_{h,i}^{V_j}(\alpha) - \rho(\alpha - \min_{s'} [V_j]_\alpha(s')) \right\}, \quad \forall i \in [d], j \in \{r, g\}, \quad (8)$$

where $\lambda_0 > 0$ is the regularization coefficient,

$$\Lambda_h = \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \phi(s_h^\tau, a_h^\tau)^\top + \lambda_0 I_d \quad (9)$$

is the cumulative sample covariance matrix, and $\bar{\nu}_{h,i}^{V_j}(\alpha)$ is the i -th coordinate of $\bar{\nu}_h^{V_j}(\alpha)$ defined by

$$\bar{\nu}_h^{V_j}(\alpha) = \arg \min_{\nu \in \mathbb{R}^d} \sum_{\tau \in \mathcal{D}_h^0} (\phi(s_h^\tau, a_h^\tau)^\top \nu - [V_j]_\alpha(s_{h+1}^\tau))^2 + \lambda_0 \|\nu\|_2^2 = \Lambda_h^{-1} \left(\sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) [V_j]_\alpha(s_{h+1}^\tau) \right).$$

Leveraging the linearity of the robust Bellman operator shown in Lemma 1, we construct the empirical robust Bellman operator $\hat{\mathbb{B}}_h^\rho$ to approximate $\mathbb{B}_h^\rho$: for any function $V_j : \mathcal{S} \rightarrow [0, H]$,

$$(\hat{\mathbb{B}}_{j,h}^\rho V_j)(s, a) = \phi(s, a)^\top (\hat{\theta}_{j,h} + \hat{\nu}_h^{\rho, V_j}), \quad \forall (s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H], \quad j \in \{r, g\}. \quad (10)$$

3.3 CROP-VI: constrained robust optimistic-pessimistic value iteration

To compute the empirical Bellman equation (10) recursively over each horizon $h \in [H]$, we propose constrained robust optimistic-pessimistic value iteration (CROP-VI), summarized in Algorithm 1.

Algorithm 1 begins by constructing a subsampled dataset $\mathcal{D}^0$ from the original batch $\mathcal{D}$ using the Two-fold-subsampling (Algorithm 2), following [60] to mitigate statistical dependencies across time steps. Further details on Two-fold-subsampling and its guarantees are provided in Section B.2. Given $\mathcal{D}_0$ and the terminal conditions $\hat{V}_{r,H+1}(\cdot) = \hat{V}_{g,H+1}(\cdot) = 0$, each iteration of CROP-VI at step h proceeds in two phases:

- **Robust Bellman construction.** Conditioned on the fixed $\hat{V}_{j,h+1}(j \in \{r, g\})$ from the previous iteration (line 4-9 in Algorithm 1), we build the empirical robust Bellman operators via (7)-(10).

- **Q -functions update.** Using the robust Bellman operators, we then estimate the pessimistic reward Q -function and optimistic constraint Q -function as follows: for all $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$\bar{Q}_{r,h}(s, a) = (\hat{\mathbb{B}}_{r,h}^\rho \hat{V}_{r,h+1})(s, a) - \Gamma_h(s, a), \quad \bar{Q}_{g,h}(s, a) = (\hat{\mathbb{B}}_{g,h}^\rho \hat{V}_{g,h+1})(s, a) + \Gamma_h(s, a)$$

where $\Gamma_h : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is a state-action-dependent penalty/bonus term defined as $\Gamma_h(s, a) := \gamma_0 \sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}}$ with $\gamma_0 > 0$ controlling the regularization strength.

Design rationale. In offline RL, the principle of pessimism is routinely invoked to compensate for the limited coverage inherent in batch data [33, 61, 62, 40]. Applying pessimism to both the reward and the constraints guarantees safety, but often at the cost of excessive conservatism. To strike a better balance, we adopt an *asymmetric* strategy: we remain pessimistic with respect to the reward function to prevent over-estimation, while we take an optimistic stance toward the constraint value. This optimism encourages exploratory behavior yet still enforces feasibility, as the update ensures $\hat{V}_g \geq b$. Although $\hat{V}_g \geq b$ does not imply $V_g^* \geq b$, our analysis guarantees that $|\hat{V}_g - V_g^*| \leq \epsilon$ with high probability, thereby tightly bounding the gap between the estimated and true constraint values.

4 Theoretical Guarantees

In this section, we establish a high-probability sub-optimality bound for the policy generated by CROP-VI and the guarantee of sample complexity under *partial* and *full* feature coverage. We further extend the analysis to accommodate model misspecification.

4.1 Main results

Theorem 1. Fix $\beta = H/\epsilon$ and consider any $\delta \in (0, 1/3)$. Suppose Assumptions 1 and 2 hold. Set $\lambda_0 = 1$, $\gamma_0 = 6\sqrt{d\xi_0}H$ in Algorithm 1, where $\xi_0 = \log(3HK/\delta)$. Then, with probability at least $1 - 3\delta$, the policy $\hat{\pi}$ generated by Algorithm 1 satisfies

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq \tilde{O}(dH^2) \max_{P \in \mathcal{P}^\rho(P^0)} \mathbb{E}_{\pi^*, P} [\|\phi_i(s_h, a_h) \mathbb{1}_i\|_{\Lambda_h^{-1}}], \quad \text{Violation}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq \epsilon. \quad (11)$$

Theorem 1 demonstrates that our method achieves ϵ -level constraint violation, while providing an instance-dependent regret bound without requiring any data coverage assumption. Notably, the regret bound is independent of the dimensionality of the state space, depending instead on the feature space

dimension. The bound is governed by the confidence parameter δ and a general complexity term reflecting how effectively the offline dataset covers the feature space. This result highlights that the regret is fundamentally determined by the quality of offline data, encapsulated by the expected feature exploration. Building on this foundation, we further analyze the sample complexity required to achieve an ϵ -regret and ϵ -violation optimal policy under varying data coverage conditions.

4.2 The case of partial feature coverage

We first consider the scenario of partial feature coverage (a.k.a. sufficient feature coverage), where the behavior policy is only required to cover all feature dimensions visited by the optimal robust policy. Formally, we consider the following clipped single-policy concentrability condition, which is used in [40] for Lin-RMDPs. This condition quantifies the worst-case discrepancy between the occupancy measure induced by the optimal robust policy π^* under any kernel $P \in \mathcal{P}^\rho(P^0)$ and that of the behavior policy π^b under the nominal kernel P^0 for every feature $i \in [d]$.

Assumption 3 (Robust single-policy clipped concentrability). *The dataset $\mathcal{D}$, collected by the behavior policy π^b , satisfies*

$$\forall (i, h, P) \in [d] \times [H] \times \mathcal{P}^\rho(P^0), \min\{\mathbb{E}_{d_h^{\pi^*, P}} \phi_i^2(s, a), 1/d\} \leq C_{\text{rob}}^*/d \cdot \lambda_{\min}(\mathbb{E}_{d_h^b}[\phi(s, a)\phi(s, a)^\top]),$$

for some finite quantity $C_{\text{rob}}^* \in [1, \infty)$. In addition, we follow the convention $0/0 = 0$.

The quantity C_{rob}^* is a clipped concentrability constant that measures the relative feature coverage of the batch data compared to an ideal dataset generated by π^* . A smaller C_{rob}^* implies better alignment with π^* , (i.e., higher data quality), while a larger C_{rob}^* indicates worse data quality. Importantly, CRDP-VI does not require prior knowledge of C_{rob}^* during implementation; it suffices to assume $C_{\text{rob}}^* < \infty$, ensuring that π^b sufficiently explores every feature dimension that π^* visits.

Under Assumption 3, we establish the following sample complexity guarantee for achieving an ϵ -optimal and ϵ -constrained robust policy.

Corollary 1 (Partial feature coverage). *Suppose Assumptions 1, 2 and 3 hold. Consider any $\delta \in (0, 1/4)$ and the same hyperparameter settings as in Theorem 1. Let $d_{\min}^b = \min_{h,s,a} \{d_h^b(s, a) : d_h^b(s, a) > 0\}$. Then, with probability exceeding $1 - 4\delta$, the ϵ -level constrained policy $\hat{\pi}$ returned by Algorithm 1 achieves*

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq \mathcal{O}\left(dH^2 \sqrt{C_{\text{rob}}^* \log(3HK/\delta)/K}\right)$$

provided that $K \geq c_0 \log(KH/\delta)/d_{\min}^b$ for some sufficiently large universal constant $c_0 > 0$. In other words, the learned ϵ -constrained policy $\hat{\pi}$ is ϵ -optimal if the total number of sample trajectories satisfies $K \geq \tilde{O}(C_{\text{rob}}^* d^2 H^4 / \epsilon^2)$.

Corollary 1 is a direct consequence of Theorem 1 and establishes a regret bound that scales with the feature dimension d and the horizon length H . This bound is comparable to those achieved in prior work on standard linear MDPs [33] and unconstrained distributionally robust linear MDPs [40].

4.3 The case of full feature coverage

We next introduce the following full feature coverage assumption, which is commonly employed in the analysis of offline linear MDPs [33, 51, 34, 50], and requires the behavior policy to uniformly cover the feature space.

Assumption 4 (Full data coverage). *We assume $\kappa = \min_{h \in [H]} \lambda_{\min}(\mathbb{E}_{d_h^b}[\phi(s, a)\phi(s, a)^\top]) > 0$.*

Compared to Assumption 3, Assumption 4 necessitates the behavior policy to be more exploratory to reach every feature dimension, which is a stronger assumption requiring full coverage of all feature dimensions. The following corollary provides the sample complexity guarantee of CRDP-VI under the full feature coverage.

Corollary 2 (Full feature coverage). *Suppose Assumptions 1, 2 and 4 hold. Consider any $\delta \in (0, 1/5)$ and the same hyperparameter settings as in Theorem 1. Let $d_{\min}^b = \min_{h,s,a} \{d_h^b(s, a) : d_h^b(s, a) >$*

0}. Then, with probability at least $1 - 5\delta$, the ϵ -constrained policy $\hat{\pi}$ returned by Algorithm 1 achieves

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq \mathcal{O}\left(H^2 \sqrt{d \log(3HK/\delta) \kappa^{-1} K^{-1}}\right),$$

provided that $K \geq \max\{c_0 \log(2Hd/\delta)/\kappa^2, c_0 \log(KH/\delta)/d_{\min}^b\}$ for some sufficiently large universal constant c_0 . In other words, the learned ϵ -constrained policy $\hat{\pi}$ is ϵ -optimal if the total number of sample trajectories satisfies $K \geq \tilde{O}\left(dH^4/\kappa^{-1}\epsilon^{-2}\right)$.

Corollary 2 is a direct consequence of Theorem 1. In contrast to the tabular case [25], which controls the finite sample error separately for each (s, a) pair, our analysis leverages the shared linear structure across all (s, a) pairs and constructs a novel ϵ -net to uniformly bound the finite sample error over the entire function class.

4.4 Extension

The soft state-aggregation assumption may be overly restrictive in practice. We therefore relax the assumption of the soft state-aggregation MDP by allowing the true transition kernel dynamics to deviate slightly from the state-aggregation transition.

Assumption 5 (Model misspecification in transition model). *For each $h \in [H]$, there exists a kernel $\tilde{P}_h^0 \in \text{Span}\{\phi(s, a)\}$ and a deviation coefficient $\psi \geq 0$ such that, for all (s, a) , the true transition kernel $P_h^0(\cdot|s, a)$ satisfies $\|P_h^0(\cdot|s, a) - \tilde{P}_h^0(\cdot|s, a)\|_1 \leq \psi$. Here, $\text{Span}\{\phi(s, a)\}$ denotes the set of all linear combinations of the feature vector $\phi(s, a)$. We continue to assume, without loss of generality, that the reward/utility functions satisfy $r_h, g_h \in \text{Span}\{\phi(s, a)\}$ for all $h \in [H]$.*

Corollary 3 (Partial feature coverage). *Suppose Assumptions 1, 2, 3 and 5 hold. Consider any $\delta \in (0, 1/4)$ and the same hyperparameter settings as in Theorem 1. Let $d_{\min}^b = \min_{h,s,a}\{d_h^b(s, a) : d_h^b(s, a) > 0\}$. Then, with probability exceeding $1 - 4\delta$, the ϵ -level constrained policy $\hat{\pi}$ returned by Algorithm 1 achieves*

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq 16dH^2 \sqrt{\frac{C_{\text{rob}}^*}{K}} \left(6 \log\left(\frac{3HK}{\delta}\right) + \psi\right) + \frac{H(H-1)\psi}{2}$$

provided that $K \geq c_0 \log(KH/\delta)/d_{\min}^b$ for some sufficiently large universal constant $c_0 > 0$.

Corollary 4 (Full feature coverage). *Suppose Assumptions 1, 2, 3 and 5 hold. Consider any $\delta \in (0, 1/5)$ and the same hyperparameter settings as in Theorem 1. Let $d_{\min}^b = \min_{h,s,a}\{d_h^b(s, a) : d_h^b(s, a) > 0\}$. Then, with probability at least $1 - 5\delta$, the ϵ -level constrained policy $\hat{\pi}$ returned by Algorithm 1 achieves*

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq \frac{16\sqrt{d}H^2}{\sqrt{\kappa K}} \left(6 \log\left(\frac{3HK}{\delta}\right) + \psi\right) + \frac{H(H-1)\psi}{2},$$

provided that $K \geq \max\{c_0 \log(2Hd/\delta)/\kappa^2, c_0 \log(KH/\delta)/d_{\min}^b\}$ for some sufficiently large universal constant c_0 .

Corollaries 3 and 4 quantify the impact of model misspecification: when the soft-state aggregation model deviates from the true one by at most ψ in total variation, the regret of the learned policy degrades by an additive factor of $\mathcal{O}(\psi H^2 \cdot (d\sqrt{C_{\text{rob}}^*}/K + 1))$ under partial feature coverage, and $\mathcal{O}(\psi H^2 \cdot (\sqrt{d\kappa^{-1}K^{-1}} + 1))$ under full feature coverage. The degradation performance depends on the divergence between the true transition distribution and the robustness level we aim to achieve.

5 Conclusion

In this paper, we investigate the sample complexity for distributional robust constraint offline RL in linear MDPs with TV uncertainty sets. To this end, we develop a distributionally robust variant of constraint optimistic-pessimistic least-squares value iteration, called CROP-VI. We establish the regret bound and the constraint violation bound for Lin-RCMDPs under various offline data assumptions. We further extend our analysis to accommodate transition model misspecification. In the future, an important direction is to explore alternative uncertainty sets and to establish the lower bound across the full range of the uncertainty levels.

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853 A Technical Lemmas

854 **Lemma 3** (Hoeffding-type inequality for self-normalized process [63]). *Let $\{\eta_t\}_{t=1}^\infty$ be a real-valued*
 855 *stochastic process and let $\{\mathcal{F}_t\}_{t=0}^\infty$ be a filtration such that η_t is $\mathcal{F}_t$ -measurable. Let $\{x_t\}_{t=1}^\infty$ be an*
 856 *$\mathbb{R}^d$ -valued stochastic process where x_t is $\mathcal{F}_{t-1}$ measurable and $x_t \leq L$. Let $\Lambda_t = \lambda I_d + \sum_{s=1}^t x_s x_s^\top$.*
 857 *Assume that conditioned on $\mathcal{F}_{t-1}$, η_t is mean-zero and R -sub-Gaussian. Then, for any $\delta > 0$, with*
 858 *probability at least $1 - \delta$, for all $t > 0$, we have*

$$\left\| \sum_{s=1}^t x_s \eta_s \right\|_{\Lambda_t^{-1}} \leq R \sqrt{d \log(1 + tL/\lambda) + 2 \log(1/\delta)}.$$

859 **Lemma 4** (Lemma 5.1 in [33]). *Under the condition that with probability at least $1 - \delta$, the penalty*
 860 *function $\Gamma_h : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ in Algorithm 1 and satisfying*

$$|(\widehat{\mathbb{B}}_{r,h}^\rho \widehat{V}_{r,h+1})(s, a) - (\mathbb{B}_{r,h}^\rho \widehat{V}_{r,h+1})(s, a)| \leq \Gamma_h(s, a), \quad \forall (s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H], \quad (12)$$

861 *we have*

$$0 \leq \iota_h(s, a) \leq 2\Gamma_h(s, a), \quad \forall (s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H].$$

862 **Lemma 5** (Lemma 7 in [40]). *For any positive semi-definite matrix $A \in \mathbb{R}^{d \times d}$ and any constant*
 863 *$c \geq 0$, we have*

$$\text{Tr}(A(I + cA)^{-1}) \leq \sum_{i=1}^d \frac{\lambda_i}{1 + c\lambda_i},$$

864 *where $\{\lambda_i\}_{i=1}^d$ are the eigenvalues of A and $\text{Tr}(\cdot)$ denotes the trace of the given matrix.*

865 **Lemma 6** (Lemma H.5 in [64]). *Let $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ be a bounded function such that $\|\phi(s, a)\|_2 \leq C$*
 866 *for all $(s, a) \in \mathcal{S} \times \mathcal{A}$. For any $K > 0$ and $\lambda > 0$, define $\bar{G}_K = \sum_{k=1}^K \phi(s_k, a_k) \phi(s_k, a_k)^\top + \lambda I_d$*
 867 *where (s_k, a_k) are i.i.d. samples from some distribution ν over $\mathcal{S} \times \mathcal{A}$. Let $G = \mathbb{E}_\nu[\phi(s, a) \phi(s, a)^\top]$.*
 868 *Then for any $\delta \in (0, 1)$, if K satisfies that*

$$K \geq \max\{512C^4 \|G^{-1}\|^2 \log(2d/\delta), 4\lambda \|G^{-1}\|\}.$$

869 *Then with probability at least $1 - \delta$, it holds simultaneously for all $u \in \mathbb{R}^d$ that*

$$\|u\|_{\bar{G}_K^{-1}} \leq \frac{2}{\sqrt{K}} \|u\|_{G^{-1}}.$$

870 **Lemma 7** (Lemma 6 in [40]). *For any function $f_1 : \mathcal{C} \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $f_2 : \mathcal{C} \subseteq \mathbb{R} \rightarrow \mathbb{R}$, we have*

$$\max_{\alpha \in \mathcal{C}} f_1(\alpha) - \max_{\alpha \in \mathcal{C}} f_2(\alpha) \leq \max_{\alpha \in \mathcal{C}} (f_1(\alpha) - f_2(\alpha)).$$

871 B Analysis for CROP-VI: Algorithm 1

872 B.1 Proof of Lemma 2

873 *Proof.* We prove this by contradiction. Assume the claim is false. Let π^* be the optimal policy for
 874 the original problem (3). Feasibility of π^* implies $(b - V_{1,g}^{\pi^*,\rho}(\zeta))_+ = 0$, hence

$$V_{r,1}^{\pi^*,\rho}(\zeta) - \lambda \left(b - V_{g,1}^{\pi^*,\rho}(\zeta) \right)_+ \geq 0, \quad (13)$$

875 as $r_h(s, a) \in [0, 1]$. For any policy π —in particular the optimizer $\widehat{\pi}$ of the rectified problem (5)—we
 876 have

$$V_{r,1}^{\pi,\rho}(\zeta) - \lambda \left(b - V_{g,1}^{\pi,\rho}(\zeta) \right)_+ \leq H - \lambda \left(b - V_{g,1}^{\pi,\rho}(\zeta) \right)_+. \quad (14)$$

877 By the contradictory assumption, $(b - V_{g,1}^{\widehat{\pi},\rho}(\zeta)) > \epsilon$. Substituting into (14) with $\lambda = H/\epsilon$ yields

$$V_{r,1}^{\widehat{\pi},\rho}(\zeta) - \lambda \left(b - V_{g,1}^{\widehat{\pi},\rho}(\zeta) \right)_+ < H - (H/\epsilon) \epsilon = 0,$$

878 which contradicts the lower bound (13). Therefore, the optimal policy of (5) violates the constraint
 879 by at most ϵ . Moreover, any infeasible policy must satisfy $b - V_{g,1}^{\pi,\rho}(\zeta) < \epsilon$, so only the feasible
 880 policies of the original robust CMDP (3) can be the optimal solution of (5). Hence,

$$V_{r,1}^{\widehat{\pi},\rho} - \lambda \left(b - V_{g,1}^{\widehat{\pi},\rho} \right)_+ \leq V_{r,1}^{\pi^*,\rho} - \lambda \left(b - V_{g,1}^{\pi^*,\rho} \right)_+,$$

881 where we have used the fact that $(b - V_{g,1}^{\pi^*,\rho})_+ = 0$, and $(b - V_{g,1}^{\widehat{\pi},\rho})_+ = 0$. Thus, the optimal solution
 882 of (3) is also optimal in (5). $\square$

Algorithm 2: Two-fold subsampling

Input: Batch dataset $\mathcal{D}$;

- 1 **Split Data:** Split $\mathcal{D}$ into two halves $\mathcal{D}^{\text{main}}$ and $\mathcal{D}^{\text{aux}}$, where $|\mathcal{D}^{\text{main}}| = |\mathcal{D}^{\text{aux}}| = N_h/2$. Denote $N_h^{\text{main}}(s)$ (resp. $N_h^{\text{aux}}(s)$) as the number of sample transitions from state s at step h in $\mathcal{D}^{\text{main}}$ (resp. $\mathcal{D}^{\text{aux}}$);
- 2 **Construct the high-probability lower bound $N_h^{\text{trim}}(s)$ by $\mathcal{D}^{\text{aux}}$:** For each $s \in \mathcal{S}$ and $1 \leq h \leq H$, compute

$$N_h^{\text{trim}}(s) = \max\{N_h^{\text{aux}}(s) - 10\sqrt{N_h^{\text{aux}}(s) \log \frac{KH}{\delta}}, 0\}. \quad (15)$$

Construct the almost temporally statistically independent $\mathcal{D}^{\text{trim}}$: Let $\mathcal{D}_h^{\text{main}}(s)$ denote the dataset containing all transition-reward sample tuples at the current state s and step h from $\mathcal{D}^{\text{main}}$. For any $(s, h) \in \mathcal{S} \times [H]$, subsample $\min\{N_h^{\text{trim}}(s), N_h^{\text{main}}(s)\}$ transition-reward sample tuples randomly from $\mathcal{D}_h^{\text{main}}(s)$, denoted as $\mathcal{D}^{\text{main,sub}}$;

Output: $\mathcal{D}^0 = \mathcal{D}^{\text{main,sub}}$.

883 B.2 Two-fold subsampling method

884 To mitigate the temporal dependency in the batch dataset $\mathcal{D}$, we adopt the two-fold subsampling idea
 885 of [60]. The key idea is to utilize half of the data to establish a valid lower bound of the number of
 886 samples, which is employed to achieve the statistical independence in the remaining half of the dataset.
 887 The detailed implementation of the two-fold subsampling can be summarized in the Algorithm 2.
 888 Recall that we assume the sample trajectories in $\mathcal{D}$ are generated independently. Then, the following
 889 lemma, adapted from Lemma 9 in [40]—itself a slight modification of Lemma 3 and Lemma 7 in
 890 [60]—establishes that (15) is a valid lower bound of $N_h^{\text{main}}(s)$ for any $s \in \mathcal{S}$ and $h \in [H]$.

891 **Lemma 8** (Lemma 9 in [40]). *With probability at least $1 - 2\delta$, if $N_h^{\text{trim}}(s)$ satisfies (15) for every
 892 $s \in \mathcal{S}$ and $h \in [H]$, then $\mathcal{D}^0 := \mathcal{D}^{\text{main,sub}}$ contains temporally statistically independent samples
 893 and the following bound holds simultaneously, i.e.,*

$$N_h^{\text{trim}}(s) \leq N_h^{\text{main}}(s), \quad \forall (s, h) \in \mathcal{S} \times [H].$$

894 *In addition, with probability at least $1 - 3\delta$, the following lower bound also holds, i.e.,*

$$N_h^{\text{trim}}(s, a) \geq \frac{K d_h^b(s, a)}{8} - 5\sqrt{K d_h^b(s, a) \log(\frac{KH}{\delta})}, \quad \forall (s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H].$$

895 B.3 Proof of Theorem 1

896 We primarily focus on the model misspecification setting. Notably, when the model misspecification
 897 parameter $\psi = 0$, we can recover the results in Theorem 1.

898 **Notations.** Before presenting the proof of Theorem 1, we introduce several notations for clarity.
 899 We define the model evaluation error at the step h of CROP-VI as

$$\iota_{r,h}(s, a) = \mathbb{B}_{r,h}^\rho \widehat{V}_{r,h+1}(s, a) - \widehat{Q}_{r,h}(s, a), \quad \forall (h, s, a) \in [H] \times \mathcal{S} \times \mathcal{A}. \quad (16)$$

900 In addition, we denote the estimated weight of the transition kernel at the h -th step by

$$\forall (s, h) \in \mathcal{S} \times [H]: \quad \widehat{\mu}_h(s) = \Lambda_h^{-1} \left(\sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \mathbb{1}(s_{h+1}^\tau = s) \right) \in \mathbb{R}^d, \quad (17)$$

901 where $\mathbb{1}(\cdot)$ is the indicator function. Accordingly, it holds that $\widehat{\nu}_h^{\widehat{V}_r}(\alpha) =$
 902 $\int_{\mathcal{S}} \widehat{\mu}_h(s') [\widehat{V}_{h+1,r}(s')]_\alpha ds' \in \mathbb{R}^d$. We denote the set of all the possible state occupancy
 903 distributions associated with the optimal policy π^* and any $P \in \mathcal{P}^\rho(P^0)$ as

$$\mathcal{D}_h^* = \left\{ \left[d_h^{*,P}(s) \right]_{s \in \mathcal{S}} : P \in \mathcal{P}^\rho(P^0) \right\} = \left\{ \left[d_h^{*,P}(s, \pi_h^*(s)) \right]_{s \in \mathcal{S}} : P \in \mathcal{P}^\rho(P^0) \right\}, \quad (18)$$

904 for any time step $h \in [H]$.

905 B.3.1 Proof sketch for Theorem 1

906 We first claim that Theorem 1 holds as long as the following theorem can be established.

907 **Theorem 2.** Consider $\delta \in (0, 1)$. Suppose that the dataset $\mathcal{D}_0$ in Algorithm 1 contains $N_h < K$
 908 transition-reward sample tuples at every $h \in [H]$. Assume that conditional on $\{N_h\}_{h \in [H]}$, all the
 909 sample tuples in $\mathcal{D}_h^0$ are statistically independent. Suppose that Assumption 1, 2, and 5 hold. In
 910 *CR0P-VI*, we set

$$\lambda_0 = 1, \quad \gamma_0 = 6\sqrt{d\xi_0}H + \sqrt{d}\psi H, \quad \text{where } \xi_0 = \log(3HK/\delta). \quad (19)$$

911 Here, $\delta \in (0, 1)$ is the confidence parameter and K is the upper bound of N_h for any $h \in [H]$. Then,
 912 $\{\hat{\pi}_h\}_{h=1}^H$ generated by Algorithm 1, with the probability at least $1 - \delta$, satisfies

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq \tilde{O}(\sqrt{d}H \cdot \max\{1, \psi\}) \sum_{h=1}^H \sum_{i=1}^d \max_{d_h^* \in \mathcal{D}_h^*} \mathbb{E}_{d_h^*} \left[\|\phi_i(s_h, a_h) \mathbb{1}_i\|_{\Lambda_h^{-1}} \right] + \frac{H(H-1)\psi}{2}.$$

913 As the construction in Algorithm 2, $\{N_h^{\text{trim}}(s)\}_{s \in \mathcal{S}, h \in [H]}$ is computed using $\mathcal{D}^{\text{aux}}$ that is independent
 914 of $\mathcal{D}^0$. From Lemma 8, $N_h^{\text{trim}}(s)$ is a valid sampling number for any $s \in \mathcal{S}$ and $h \in [H]$ such
 915 that $|\mathcal{D}_h^0| = \sum_{s \in \mathcal{S}} N_h^{\text{trim}}(s) \leq K$, and $\mathcal{D}_h^0$ can be treated as containing temporally statistically
 916 independent samples with probability exceeding $1 - 2\delta$. Therefore, by invoking Theorem 2 with
 917 $N_h := |\mathcal{D}_h^0|$, we have

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq \tilde{O}(\sqrt{d}H \cdot \max\{1, \psi\}) \sum_{h=1}^H \sum_{i=1}^d \max_{d_h^* \in \mathcal{D}_h^*} \mathbb{E}_{d_h^*} \left[\|\phi_i(s_h, a_h) \mathbb{1}_i\|_{\Lambda_h^{-1}} \right] + \frac{H(H-1)\psi}{2},$$

918 with probability exceeding $1 - 3\delta$.

919 B.3.2 Proof of Theorem 2

920 **The sub-optimality bound.** The argument unfolds in three steps.

921 **Step 1: establishing the pessimistic property of the reward function and the optimistic property**
 922 **of the utility function.** We first substantiate the pessimism of the reward function, which heavily
 923 depends on the following lemmas.

924 **Lemma 9** (Adapted from Lemma 10 in [40]). Suppose all the assumptions in Theorem 2 hold and
 925 follow all the parameters setting in (19). Then for any $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$, with probability at
 926 least $1 - \delta$, the value/utility function $\{\hat{V}_{j,h}\}_{h=1}^H$ for $j = r, g$ generated by *CR0P-VI* satisfies

$$|(\hat{\mathbb{B}}_{j,h}^\rho \hat{V}_{j,h+1})(s, a) - (\mathbb{B}_{j,h}^\rho \hat{V}_{j,h+1})(s, a)| \leq \Gamma_h(s, a) := \gamma_0 \sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}} + (H-h)\psi. \quad (20)$$

927 In the following, we will show that the following relations hold:

$$Q_{r,h}^{*,\rho}(s, a) \geq Q_{r,h}^{\hat{\pi},\rho}(s, a) \geq \hat{Q}_{r,h}(s, a) \quad \text{and} \quad V_{r,h}^{*,\rho}(s) \geq V_{r,h}^{\hat{\pi},\rho}(s) \geq \hat{V}_{r,h}(s), \quad (21)$$

928 for all $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$, if the condition (20) holds. It implies that $\hat{Q}_{r,h}(s, a)$ and $\hat{V}_{r,h}(s)$
 929 is the pessimistic estimates of $Q_{r,h}^{\hat{\pi},\rho}(s, a)$ and $V_{r,h}^{\hat{\pi},\rho}(s)$ for any $s \in \mathcal{S}$, respectively. Notice that if
 930 $Q_{r,h}^{\hat{\pi},\rho}(s, a) \geq \hat{Q}_{r,h}(s, a)$ for all $(s, a) \in \mathcal{S} \times \mathcal{A}$, one simultaneously has the following relation:

$$V_{r,h}^{\hat{\pi},\rho}(s) = Q_{r,h}^{\hat{\pi},\rho}(s, \hat{\pi}_h(s)) \geq \hat{Q}_{r,h}(s, \hat{\pi}_h(s)) = \hat{V}_{r,h}(s), \quad \forall (s, h) \in \mathcal{S} \times [H].$$

931 Therefore, we shall verify that

$$Q_{r,h}^{\hat{\pi},\rho}(s, a) \geq \hat{Q}_{r,h}(s, a), \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A}, \quad (22)$$

932 by induction, and $V_{r,h}^{\hat{\pi},\rho}(s) \geq \hat{V}_{r,h}(s)$ will spontaneously hold for $s \in \mathcal{S}$.

933 • At step $h = H + 1$: From the initialization step in Algorithm 1, we have $Q_{r,H+1}^{\hat{\pi},\rho}(s, a) =$
 934 $\hat{Q}_{r,H+1}(s, a) = 0$, for any $(s, a) \in \mathcal{S} \times \mathcal{A}$, and (22) holds.

935 • For any step $h \leq H$: Suppose $Q_{r,h+1}^{\hat{\pi},\rho}(s,a) \geq \hat{Q}_{r,h+1}(s,a)$. From (21), we have
 936 $V_{r,h+1}^{\hat{\pi},\rho}(s) \geq \hat{V}_{r,h+1}(s)$. Therefore, if $\hat{Q}_{r,h}(s,a) = 0$, $Q_{r,h}^{\hat{\pi},\rho}(s,a) \geq 0 = \hat{Q}_{r,h}(s,a)$,
 937 for any $(s,a) \in \mathcal{S} \times \mathcal{A}$. Otherwise,

$$\begin{aligned}\hat{Q}_{r,h}(s,a) &\leq (\mathbb{B}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) - \Gamma_h(s,a) \\ &= (\mathbb{B}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) + (\hat{\mathbb{B}}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) - (\mathbb{B}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) - \Gamma_h(s,a) \\ &\leq (\mathbb{B}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) + |(\hat{\mathbb{B}}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) - (\mathbb{B}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a)| - \Gamma_h(s,a) \\ &\leq (\mathbb{B}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) + \Gamma_h(s,a) - \Gamma_h(s,a) \\ &\leq (\mathbb{B}_{r,h}^{\rho} \hat{V}_{r,h+1})(s,a) \\ &\leq (\mathbb{B}_{r,h}^{\rho} V_{r,h+1}^{\hat{\pi},\rho})(s,a) = Q_{r,h}^{\hat{\pi},\rho}(s,a),\end{aligned}$$

938 where the first inequality is from the definition of $\hat{Q}_{r,h}(s,a)$ (cf. Line 14 in Algorithm 1),
 939 and third inequality is based on the condition (20).

940 Combining these two cases, for any $h \in [H+1]$, we could verify the pessimistic property, i.e. the
 941 equation (21).

942 Next, we establish the optimism of the utility function, leveraging Lemma 9. Building on this result,
 943 we will demonstrate that the following relations hold:

$$Q_{g,h}^{\hat{\pi},\rho}(s,a) \leq \hat{Q}_{g,h}(s,a) \quad \text{and} \quad V_{g,h}^{\hat{\pi},\rho}(s) \leq \hat{V}_{g,h}(s), \quad \forall (s,a,h) \in \mathcal{S} \times \mathcal{A} \times [H], \quad (23)$$

944 if the condition (20) holds. It implies that $\hat{Q}_{g,h}(s,a)$ and $\hat{V}_{g,h}(s)$ is the optimistic estimates of
 945 $Q_{g,h}^{\hat{\pi},\rho}(s,a)$ and $V_{g,h}^{\hat{\pi},\rho}(s)$ for any $s \in \mathcal{S}$, respectively. Notice that if $Q_{g,h}^{\hat{\pi},\rho}(s,a) \leq \hat{Q}_{g,h}(s,a)$ for all
 946 $(s,a) \in \mathcal{S} \times \mathcal{A}$, one simultaneously has the following relation:

$$V_{g,h}^{\hat{\pi},\rho}(s) = Q_{g,h}^{\hat{\pi},\rho}(s, \hat{\pi}_h(s)) \leq \hat{Q}_{g,h}(s, \hat{\pi}_h(s)) = \hat{V}_{g,h}(s), \quad \forall (s,h) \in \mathcal{S} \times [H].$$

947 Therefore, we shall verify that

$$Q_{g,h}^{\hat{\pi},\rho}(s,a) \leq \hat{Q}_{g,h}(s,a), \quad \forall (s,a) \in \mathcal{S} \times \mathcal{A}, \quad (24)$$

948 by induction, and $V_{g,h}^{\hat{\pi},\rho}(s) \leq \hat{V}_{g,h}(s)$ will spontaneously hold for $s \in \mathcal{S}$.

949 • At step $h = H+1$: From the initialization step in Algorithm 1, we have $Q_{g,H+1}^{\hat{\pi},\rho}(s,a) =$
 950 $\hat{Q}_{g,H+1}(s,a) = 0$, for any $(s,a) \in \mathcal{S} \times \mathcal{A}$, and (24) holds.
 951 • For any step $h \leq H$: Suppose $Q_{g,h+1}^{\hat{\pi},\rho}(s,a) \leq \hat{Q}_{g,h+1}(s,a)$. From (23), we have
 952 $V_{g,h+1}^{\hat{\pi},\rho}(s) \leq \hat{V}_{g,h+1}(s)$. Therefore, if $\hat{Q}_{g,h}(s,a) = H-h+1$, $Q_{g,h}^{\hat{\pi},\rho}(s,a) \leq H-h+1 =$
 953 $\hat{Q}_{g,h}(s,a)$, for any $(s,a) \in \mathcal{S} \times \mathcal{A}$. Otherwise,

$$\begin{aligned}Q_{g,h}^{\hat{\pi},\rho}(s,a) &= (\mathbb{B}_{g,h}^{\rho} V_{g,h+1}^{\hat{\pi},\rho})(s,a) \\ &\leq (\mathbb{B}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a) \\ &\leq (\hat{\mathbb{B}}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a) + (\mathbb{B}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a) - (\hat{\mathbb{B}}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a) \\ &\leq (\hat{\mathbb{B}}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a) + |(\mathbb{B}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a) - (\hat{\mathbb{B}}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a)| \\ &\leq (\mathbb{B}_{g,h}^{\rho} \hat{V}_{g,h+1})(s,a) + \Gamma_h(s,a) = \hat{Q}_{g,h}(s,a),\end{aligned}$$

954 where the first inequality is from the definition of $\hat{Q}_{g,h}(s,a)$ (cf. Line 15 in Algorithm 1),
 955 and third inequality is based on the condition (20).

956 Combining these two cases, for any $h \in [H+1]$, we could verify the optimistic property, i.e. the
 957 equation (23).

958 **Step 2: bounding the suboptimality gap.** Notice that for any $h \in [H]$ and any $s \in \mathcal{S}$,

$$V_{r,h}^{\star,\rho}(s) - V_{r,h}^{\widehat{\pi},\rho}(s) = \underbrace{V_{r,h}^{\star,\rho}(s) - \widehat{V}_{r,h}^{\pi^\star,\rho}(s)}_{T_1} + \underbrace{\widehat{V}_{r,h}^{\pi^\star,\rho}(s) - \widehat{V}_{r,h}^{\widehat{\pi},\rho}(s)}_{T_2} + \underbrace{\widehat{V}_{r,h}^{\widehat{\pi},\rho}(s) - V_{r,h}^{\widehat{\pi},\rho}(s)}_{T_3}.$$

959 We first control T_1 . For any $s \in \mathcal{S}$,

$$V_{r,h}^{\star,\rho}(s) - \widehat{V}_{r,h}^{\pi^\star,\rho}(s) = Q_{r,h}^{\star,\rho}(s, \pi^\star(s)) - \widehat{Q}_{r,h}^{\pi^\star}(s, \pi^\star(s)). \quad (25)$$

960 From the definition of the model evaluation error (i.e., equation (16)) and the robust Bellman
961 optimality equation, we have

$$\begin{aligned} \widehat{Q}_{r,h}(s, a) &= (\mathbb{B}_{r,h}^\rho \widehat{V}_{r,h+1})(s, a) - \iota_{r,h}(s, a), & \forall (s, a) \in \mathcal{S} \times \mathcal{A}, \\ Q_{r,h}^{\star,\rho}(s, a) &= (\mathbb{B}_{r,h}^\rho V_{r,h+1}^{\star,\rho})(s, a) & \forall (s, a) \in \mathcal{S} \times \mathcal{A}, \end{aligned}$$

962 which leads to

$$\begin{aligned} &Q_{r,h}^{\star,\rho}(s, \pi_h^\star(s)) - \widehat{Q}_{r,h}(s, \pi_h^\star(s)) \\ &= (\mathbb{P}_h^\rho V_{r,h+1}^{\star,\rho})(s, \pi_h^\star(s)) - (\mathbb{P}_h^\rho \widehat{V}_{r,h+1})(s, \pi_h^\star(s)) + \iota_{r,h}(s, \pi_h^\star(s)), \quad \forall s \in \mathcal{S}. \end{aligned} \quad (26)$$

963 Denote

$$P_{h,s,\pi_h^\star(s)}^{\text{inf}, \widehat{V}_r}(\cdot) := \arg \min_{P(\cdot) \in \mathcal{P}^\rho(P_{h,s,\pi_h^\star(s)}^0)} \int_{\mathcal{S}} P(s') \widehat{V}_{r,h+1}(s') ds'.$$

964 Therefore, (26) becomes

$$\begin{aligned} &Q_{r,h}^{\star,\rho}(s, \pi_h^\star(s)) - \widehat{Q}_{r,h}(s, \pi_h^\star(s)) \\ &\leq \int_{\mathcal{S}} P_{h,s,\pi_h^\star(s)}^{\text{inf}, \widehat{V}_r}(s') \left(V_{r,h+1}^{\star,\rho}(s') - \widehat{V}_{r,h+1}(s') \right) ds' + \iota_{r,h}(s, \pi_h^\star(s)), \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A}. \end{aligned} \quad (27)$$

965 Substituting (27) into (28), one has

$$V_{r,h}^{\pi^\star,\rho}(s) - \widehat{V}_{r,h}^{\widehat{\pi},\rho}(s) \leq \int_{\mathcal{S}} P_{h,s,\pi_h^\star(s)}^{\text{inf}, \widehat{V}_r}(s') \left(V_{r,h+1}^{\star,\rho}(s') - \widehat{V}_{r,h+1}(s') \right) ds' + \iota_{r,h}(s, \pi_h^\star(s)).$$

966 For any $h \in [H]$, define $\widehat{P}_{h,s}^{\text{inf}} : \mathcal{S} \rightarrow \mathcal{S}$ and $\iota_{r,h}^\star \in \mathcal{S} \rightarrow \mathbb{R}$ by

$$\widehat{P}_h^{\text{inf}}(s) = P_{h,s,\pi_h^\star(s)}^{\text{inf}, \widehat{V}_r}(\cdot) \quad \text{and} \quad \iota_{r,h}^\star(s) := \iota_{r,h}(s, \pi_h^\star(s)), \quad \forall s \in \mathcal{S}.$$

967 By telescoping sum, we finally obtain that for any $s \in \mathcal{S}$,

$$\begin{aligned} V_{r,h}^{\star,\rho}(s) - \widehat{V}_{r,h}(s) &= \langle \mathbb{1}_s, V_{r,h}^{\star,\rho} - \widehat{V}_{r,h} \rangle \\ &\leq \left(\prod_{j=h}^H \widehat{P}_j^{\text{inf}} \right) \left(V_{r,H+1}^{\star,\rho} - \widehat{V}_{r,H+1} \right)(s) + \sum_{t=h}^H \left(\prod_{j=h}^{t-1} \widehat{P}_j^{\text{inf}} \right) \iota_{r,t}^\star(s) \\ &= \sum_{t=h}^H \left(\prod_{j=h}^{t-1} \widehat{P}_j^{\text{inf}} \right) \iota_{r,t}^\star(s), \end{aligned}$$

968 where the equality is from $V_{r,H+1}^{\star,\rho}(s) = \widehat{V}_{r,H+1}(s) = 0$ and $\left(\prod_{j=t}^{t-1} \widehat{P}_j^{\text{inf}} \right)(s) = \mathbb{1}_s$.

969 We now bound the term T_2 which we describe next.

$$\begin{aligned} T_2 &= \widehat{Q}_{r,h}(s, \pi_h^\star(s)) - \widehat{Q}_{r,h}(s, \widehat{\pi}_h(s)) \\ &\leq \widehat{Q}_{r,h}(s, \pi_h^\star(s)) - \lambda \left(b - \widehat{Q}_{g,h}(s, \pi_h^\star(s)) \right)_+ - \left[\widehat{Q}_{r,h}(s, \widehat{\pi}_h(s)) - \lambda \left(b - \widehat{Q}_{g,h}(s, \pi_h^\star(s)) \right)_+ \right] \\ &\leq \widehat{Q}_{r,h}(s, \widehat{\pi}_h(s)) - \lambda \left(b - \widehat{Q}_{g,h}(s, \widehat{\pi}_h(s)) \right)_+ - \left[\widehat{Q}_{r,h}(s, \widehat{\pi}_h(s)) - \lambda \left(b - \widehat{Q}_{g,h}(s, \pi_h^\star(s)) \right)_+ \right] \\ &\leq \lambda \left(b - \widehat{Q}_{g,h}(s, \pi_h^\star(s)) \right)_+ - \lambda \left(b - \widehat{Q}_{g,h}(s, \widehat{\pi}_h(s)) \right)_+. \end{aligned}$$

Thus, if the optimism holds constraints, i.e., $\widehat{Q}_{g,h}(s, \pi^*(s)) \geq Q_{g,h}(s, \pi^*(s))$, then

$$T_2 \leq \lambda (b - Q_{g,h}(s, \pi^*(s)))_+ - \lambda (b - \widehat{Q}_{g,h}(s, \widehat{\pi}(s)))_+ \leq -\lambda (b - \widehat{Q}_{g,h}(s, \widehat{\pi}(s)))_+ \leq 0.$$

Then, we bound the term T_3 . We have

$$\widehat{V}_{r,h}^{\widehat{\pi},\rho}(s) - V_{r,h}^{\widehat{\pi},\rho}(s) = \widehat{Q}_{r,h}(s, \widehat{\pi}(s)) - Q_{r,h}(s, \widehat{\pi}(s)) \leq 0, \quad \forall s \in \mathcal{S}, \quad (28)$$

where the inequality follows from the pessimistic property of $Q_{r,h}^{\widehat{\pi},\rho}(s, a)$.

Step 3: finishing up. For any $d_h^* \in \mathcal{D}_h^*$, denote

$$d_{h:t}^* = d_h^* \left(\prod_{j=h}^{t-1} \widehat{P}_j^{\text{inf}} \right) \in \mathcal{D}_t^*.$$

The sub-optimality gap defined in (4) satisfies

$$\text{Regret}(\widehat{\pi}; \zeta, \mathcal{P}^\rho) \leq \mathbb{E}_{s_1 \sim \zeta} V_{r,1}^{\star,\rho}(s_1) - \mathbb{E}_{s_1 \sim \zeta} \widehat{V}_{r,1}(s_1) \leq \sum_{t=1}^H \mathbb{E}_{s_t \sim d_{1:t}^*} \iota_t^*(s_t) + \frac{(H-1)H\psi}{2}.$$

For any $h \in [H]$, we let $\Gamma_h^* : \mathcal{S} \rightarrow \mathbb{R}$ satisfy

$$\Gamma_h^*(s) = \Gamma_h(s, \pi_h^*(s)), \quad \forall s \in \mathcal{S}.$$

Combining Lemma 4 together with Lemma 9 will lead to

$$\text{Regret}(\widehat{\pi}; \zeta, \mathcal{P}^\rho) \leq 2 \sum_{h=1}^H \mathbb{E}_{s_h \sim d_{1:h}^*} \Gamma_h^*(s_h) + \frac{(H-1)H\psi}{2}.$$

Note that $\Gamma_h^*(s) = \gamma_0 \sum_{i=1}^d \|\phi_i(s, \pi_h^*(s))\|_{\Lambda_h^{-1}}$ for any $(s, h) \in \mathcal{S} \times [H]$. Following the definition (18), we have $d_{1:h}^* \in \mathcal{D}_h^*$ and correspondingly,

$$\begin{aligned} \text{Regret}(\widehat{\pi}; \zeta, \mathcal{P}^\rho) &\leq 2 \sum_{h=1}^H \mathbb{E}_{s_h \sim d_{1:h}^*} \Gamma_h^*(s_h) + \frac{(H-1)H\psi}{2} \\ &\leq 2\gamma_0 \sum_{h=1}^H \max_{d_h^* \in \mathcal{D}_h^*} \mathbb{E}_{(s_h, a_h) \sim d_h^*} \left[\sum_{i=1}^d \|\phi_i(s_h, a_h)\|_{\Lambda_h^{-1}} \right] + \frac{(H-1)H\psi}{2} \\ &\leq 2\gamma_0 \sum_{h=1}^H \sum_{i=1}^d \max_{d_h^* \in \mathcal{D}_h^*} \mathbb{E}_{d_h^*} \left[\|\phi_i(s_h, a_h)\|_{\Lambda_h^{-1}} \right] + \frac{(H-1)H\psi}{2}. \end{aligned}$$

with probability exceeding $1 - \delta$.

Violation bound. Now, we show the violation bound. We prove this by contradiction. Suppose that the statement is not true. Consider π^* the optimal solution for the original problem (3). Since this is feasible, then $(b - V_{1,g}^{\pi^*,\rho}(\zeta))_+ = 0$, then

$$V_{r,1}^{\pi^*,\rho}(\zeta) - \lambda (b - V_{g,1}^{\pi^*,\rho}(\zeta))_+ \geq 0, \quad (29)$$

as $r_h(s, a) \in [0, 1]$. For any π (including the optimal policy $\widehat{\pi}$) of (5), we have

$$V_{r,1}^{\pi,\rho}(\zeta) - \lambda (b - V_{g,1}^{\pi,\rho}(\zeta))_+ \leq H - \lambda (b - V_{g,1}^{\pi,\rho}(\zeta))_+. \quad (30)$$

For the optimal policy $\widehat{\pi}$, we have $(b - V_{g,1}^{\widehat{\pi},\rho}(\zeta)) > \epsilon$ (by contradiction). Hence, by (30), we have

$$V_{r,1}^{\widehat{\pi},\rho}(\zeta) - \lambda (b - V_{g,1}^{\widehat{\pi},\rho}(\zeta))_+ < H - (H/\epsilon)\epsilon = 0.$$

However, it contradicts (29) as π^* can achieve a better value for the objective in (5).

986 B.3.3 Proof of Lemma 9

987 To control $|(\widehat{\mathbb{B}}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a) - (\mathbb{B}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a)|$ for any $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$ and $j \in \{r, g\}$,
 988 we first show the following lemma, where the proof can be found in Appendix B.3.4.

989 **Lemma 10** (Adapted from Lemma 11 in [40]). *Suppose Assumption 1, 2 and 5 hold. Then, for any*
 990 *$(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$, the estimated value/utility function $\widehat{V}_{j,h+1}(j = r, g)$ generated by CROP-VI*
 991 *satisfies*

$$\begin{aligned}
 & |(\widehat{\mathbb{B}}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a) - (\mathbb{B}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a)| \\
 & \leq \left(\sqrt{\lambda_0 d} H + \sqrt{d} \xi H \right) \sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}} + (H - h) \xi \\
 & \quad + \underbrace{\max_{\alpha \in [\min_s \widehat{V}_{j,h+1}(s), \max_s \widehat{V}_{j,h+1}(s)]} \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}}}_{T_{4,h}} \sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}},
 \end{aligned} \tag{31}$$

992 where $\epsilon_h^\tau(\alpha, V_j) = \int_{\mathcal{S}} P_h^0(s' | s_h^\tau, a_h^\tau) [V_j]_\alpha(s') ds' - [V_j]_\alpha(s_{h+1}^\tau)$ for any value/utility function $V_j : \mathcal{S} \rightarrow [0, H]$, $\alpha \in [\min_s V_j(s), \max_s V_j(s)]$ and $\tau \in \mathcal{D}_h^0$.

994 We observe that the second term (i.e., $T_{4,h}$) in (31) will become dominating, as long as λ_0 and ψ are
 995 sufficiently small. In the following analysis, we will control $T_{4,h}$ via uniform concentration and the
 996 concentration of the self-normalized process.

997 Notice that α and $\widehat{V}_{j,h+1}$ are coupled with each other, which makes controlling $T_{4,h}$ intractable. To
 998 this end, we propose the minimal ϵ_0 -covering set for α . Since $\widehat{V}_{j,h+1}(s) \in [0, H]$ for any $s \in \mathcal{S}$, we
 999 construct $\mathcal{N}(\epsilon_0, H)$ as the minimal ϵ_0 -cover of the $[0, H]$ whose size satisfies $|\mathcal{N}(\epsilon_0, H)| \leq \frac{3H}{\epsilon_0}$. In
 1000 other words, for any $\alpha \in [0, H]$, there exists $\alpha^\dagger \in \mathcal{N}(\epsilon_0, H)$, we have

$$|\alpha - \alpha^\dagger| \leq \epsilon_0.$$

1001 Then we can rewrite $T_{4,h}^2$ as

$$\begin{aligned}
 T_{4,h}^2 &= \max_{\alpha \in [\min_s \widehat{V}_{j,h+1}(s), \max_s \widehat{V}_{j,h+1}(s)]} \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \left(\epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) - \epsilon_h^\tau(\alpha^\dagger, \widehat{V}_{j,h+1}) + \epsilon_h^\tau(\alpha^\dagger, \widehat{V}_{j,h+1}) \right) \right\|_{\Lambda_h^{-1}}^2 \\
 &\leq \max_{\alpha \in [0, H]} 2 \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \left(\epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) - \epsilon_h^\tau(\alpha^\dagger, \widehat{V}_{j,h+1}) \right) \right\|_{\Lambda_h^{-1}}^2 + 2 \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha^\dagger, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}}^2 \\
 &\leq 8\epsilon_0^2 K^2 / \lambda_0 + 2 \underbrace{\left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha^\dagger, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}}^2}_{T_{5,h}},
 \end{aligned} \tag{32}$$

1002 for some $\alpha^\dagger \in \mathcal{N}(\epsilon_0, H)$, where the proof of the last inequality is postponed to Appendix B.3.5.
 1003 Alternatively,

$$T_{5,h} \leq \sup_{\alpha \in \mathcal{N}(\epsilon_0, H)} \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}}^2. \tag{33}$$

1004 Noted that the samples in $\mathcal{D}_h^0$ are temporally statistically independent, i.e., $\widehat{V}_{j,h+1}$ is independent of
 1005 $\mathcal{D}_h^0$, or to say, $\widehat{\mu}_h$. Therefore, we can directly control $T_{5,h}$ via the following lemma.

1006 **Lemma 11** (Concentration of self-normalized process, adapted from Lemma 12 in [40]). *Let*
 1007 *$V_j : \mathcal{S} \rightarrow [0, H]$ be any fixed vector that is independent of $\widehat{\mu}_h$ and $\alpha \in [0, H]$ be a fixed constant.*
 1008 *For any fixed $h \in [H]$ and any $\delta \in (0, 1)$, we have*

$$P_{\mathcal{D}} \left(\left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha, V_j) \right\|_{\Lambda_h^{-1}}^2 > H^2 (2 \cdot \log(1/\delta) + d \cdot \log(1 + N_h/\lambda_0)) \right) \leq \delta.$$

1009 The proof of Lemma 11 is postponed to Appendix B.3.6. Then applying Lemma 11 and the union
 1010 bound over $\mathcal{N}(\epsilon_0, H)$, we have

$$P_{\mathcal{D}} \left(\sup_{\alpha \in \mathcal{N}(\epsilon_0, H)} \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}}^2 \geq H^2(2 \log(H|\mathcal{N}(\epsilon_0, H)|/\delta) + d \log(1 + N_h/\lambda_0)) \right) \leq \delta/H,$$

1011 for any fixed $h \in [H]$. According to [65], one has $|\mathcal{N}(\epsilon_0, H)| \leq \frac{3H}{\epsilon_0}$. Taking the union bound for
 1012 any $h \in [H]$, we arrive at

$$\sup_{\alpha \in \mathcal{N}(\epsilon_0, H)} \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}}^2 \leq 2H^2 \log\left(\frac{3H^2}{\epsilon_0 \delta}\right) + H^2 d \log\left(1 + \frac{K}{\lambda_0}\right), \quad (34)$$

1013 with probability exceeding $1 - \delta$, where we utilize $N_h \leq K$ for every $h \in [H]$ on the right-hand side.
 1014 Combining (32), (33) and (34), we have

$$T_{4,h}^2 \leq 8\epsilon_0^2 K^2 / \lambda_0 + 4H^2 \log\left(\frac{3H^2}{\epsilon_0 \delta}\right) + 2H^2 d \log\left(1 + \frac{K}{\lambda_0}\right),$$

1015 with probability at least $1 - \delta$. Let $\epsilon_0 = H/K$ and $\lambda_0 = 1$. Then,

$$T_{4,h}^2 \leq 8H^2 + 4H^2 \log\left(\frac{3HK}{\delta}\right) + 2H^2 d \log(1 + K) \leq 8H^2 + 4H^2 \log(3HK/\delta) + 2dH^2 \log(2K).$$

1016 Let $\xi_0 = \log(3HK/\delta) \geq 1$. Note that $\log(2K) \leq \log(3HK/\delta) = \xi_0$. Then, we have

$$T_{4,h}^2 \leq 8H^2 + 4H^2 \xi_0 + 2dH^2 \xi_0 \leq 16dH^2 \xi_0.$$

1017 Therefore, with probability exceeding $1 - \delta$, one has

$$\begin{aligned} & |(\widehat{\mathbb{B}}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a) - (\mathbb{B}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a)| \\ & \leq \left(\sqrt{d}H + 4\sqrt{d\xi_0}H + \sqrt{d}\psi H \right) \sum_{i=1}^d \|\phi_i(s, a)\|_{\Lambda_h^{-1}} + (H - h)\psi \\ & \leq \gamma_0 \sum_{i=1}^d \|\phi_i(s_h, a_h)\|_{\Lambda_h^{-1}} + (H - h)\psi = \Gamma_h(s, a), \end{aligned}$$

1018 where $\gamma_0 = 6\sqrt{d}H(\xi_0 + \psi)$ and the above inequality satisfies (12).

1019 B.3.4 Proof of Lemma 10

1020 *Proof.* From the DRO Bellman optimality equation, we denote

$$\begin{aligned} & (\mathbb{B}_{j,h} \widehat{V}_{j,h+1})(s, a) \\ & = j_h(s, a) + \inf_{P_{h+1} \in \mathcal{P}^\rho(P_{h+1}^0)} \mathbb{E}_{P_{h+1}(\cdot|s,a)}[\widehat{V}_{j,h+1}(s')] \\ & = j_h(s, a) + \inf_{\tilde{P}_{h+1} \in \mathcal{P}^\rho(\tilde{P}_{h+1}^0)} \mathbb{E}_{\tilde{P}_{h+1}(\cdot|s,a)}[\widehat{V}_{j,h+1}(s')] \\ & \quad + \left(\inf_{P_{h+1} \in \mathcal{P}^\rho(P_{h+1}^0)} \mathbb{E}_{P_{h+1}(\cdot|s,a)}[\widehat{V}_{j,h+1}(s')] - \inf_{\tilde{P}_{h+1} \in \mathcal{P}^\rho(\tilde{P}_{h+1}^0)} \mathbb{E}_{\tilde{P}_{h+1}(\cdot|s,a)}[\widehat{V}_{j,h+1}(s')] \right) \\ & = \phi(s, a)^\top \theta_{j,h} \\ & \quad + \sum_{i=1}^d \phi_i(s, a) \max_{\alpha \in [\min_s \widehat{V}_{j,h+1}(s), \max_s \widehat{V}_{j,h+1}(s)]} \int_{\mathcal{S}} \mu_{h,i}^0(s') [\widehat{V}_{j,h+1}]_\alpha(s') ds' - \rho(\alpha - \min_{s'} [\widehat{V}_{j,h+1}]_\alpha(s')) \\ & \quad + (H - h)\xi. \end{aligned}$$

1021 Combined with the empirical Bellman operator in our algorithm,

$$(\mathbb{B}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a) - (\widehat{\mathbb{B}}_{j,h}^\rho \widehat{V}_{j,h+1})(s, a) = \phi(s, a)^\top (\theta_{j,h} - \widehat{\theta}_{j,h}) + \phi(s, a)^\top (\widehat{\nu}_h^{\rho, \widehat{V}_j} - \nu_h^{\rho, \widehat{V}_j}) + (H - h)\psi.$$

1022 **Step 1:** We analyze the error in the reward estimation, i.e., $\phi(s, a)^\top (\theta_{j,h} - \widehat{\theta}_{j,h})$.

$$\begin{aligned} \phi(s, a)^\top (\theta_{j,h} - \widehat{\theta}_{j,h}) &= \phi(s, a)^\top \Lambda_h^{-1} \Lambda_h \theta_{j,h} - \phi(s, a)^\top \Lambda_h^{-1} \left[\sum_{\tau=1}^N \phi(s_h^\tau, a_h^\tau) j_h(s_h^\tau, a_h^\tau) \right] \\ &= \phi(s, a)^\top \Lambda_h^{-1} \Lambda_h \theta_{j,h} - \phi(s, a)^\top \Lambda_h^{-1} \left[\sum_{\tau=1}^N \phi(s_h^\tau, a_h^\tau) \phi(s_h^\tau, a_h^\tau)^\top \theta_{j,h} \right] \\ &= \phi(s, a)^\top \Lambda_h^{-1} \Lambda_h \theta_{j,h} - \phi(s, a)^\top \Lambda_h^{-1} (\Lambda_h - \lambda_0 I) \theta_{j,h} \\ &= \lambda_0 \phi(s, a)^\top \Lambda_h^{-1} \theta_{j,h} \\ &\leq \lambda_0 \|\theta_{j,h}\|_{\Lambda_h^{-1}} \|\phi(s, a)\|_{\Lambda_h^{-1}} \\ &\leq \sqrt{d\lambda_0} \|\phi(s, a)\|_{\Lambda_h^{-1}} \\ &\leq \sqrt{d\lambda_0} \sum_{i=1}^d \|\phi_i(s, a) \mathbf{1}_i\|_{\Lambda_h^{-1}}, \end{aligned}$$

1023 where the last inequality is from

$$\|\theta_{j,h}\|_{\Lambda_h^{-1}} = \sqrt{\theta_{j,h}^\top \Lambda_h^{-1} \theta_{j,h}} \leq \|\Lambda_h^{-1}\|^{1/2} \|\theta_{j,h}\| \leq \sqrt{d/\lambda_0},$$

1024 by using the fact that $\|\Lambda_h^{-1}\| \leq \lambda_0^{-1}$.

Step 2: We turn to the estimation error from the transition model, i.e., $\widehat{\nu}_h^{\rho, \widehat{V}_j} - \nu_h^{\rho, \widehat{V}_j}$. We define two auxiliary functions:

$$\widehat{g}_{h,i}(\alpha; j) = \int_{\mathcal{S}} \widehat{\mu}_{h,i}(s') [\widehat{V}_{j,h+1}]_\alpha(s') ds' - \rho(\alpha - \min_{s'} [\widehat{V}_{j,h+1}]_\alpha(s')),$$

and

$$g_{h,i}^0(\alpha; j) = \int_{\mathcal{S}} \mu_{h,i}^0(s') [\widehat{V}_{j,h+1}]_\alpha(s') ds' - \rho(\alpha - \min_{s'} [\widehat{V}_{j,h+1}]_\alpha(s')).$$

1025 Then

$$\begin{aligned} &\left| \sum_{i \in [d]} \phi_i(s, a) (\widehat{\nu}_{h,i}^{\rho, \widehat{V}_j} - \nu_{h,i}^{\rho, \widehat{V}_j}) \right| \\ &\leq \sum_{i \in [d]} \left| \phi_i(s, a) (\widehat{\nu}_{h,i}^{\rho, \widehat{V}_j} - \nu_{h,i}^{\rho, \widehat{V}_j}) \right| \\ &= \sum_{i \in [d]} \phi_i(s, a) \max_{\alpha \in [\min_s \widehat{V}_{j,h+1}(s), \max_s \widehat{V}_{j,h+1}(s)]} |\widehat{g}_{h,i}(\alpha; j) - g_{h,i}^0(\alpha; j)| \\ &\leq \sum_{i \in [d]} \phi_i(s, a) \max_{\alpha \in [\min_s \widehat{V}_{j,h+1}(s), \max_s \widehat{V}_{j,h+1}(s)]} \left| \int_{\mathcal{S}} (\widehat{\mu}_{h,i}(s') - \mu_{h,i}^0(s')) [\widehat{V}_{j,h+1}]_\alpha(s') ds' \right| \\ &= \sum_{i \in [d]} \max_{\alpha \in [\min_s \widehat{V}_{j,h+1}(s), \max_s \widehat{V}_{j,h+1}(s)]} \left| \phi_i(s, a) \int_{\mathcal{S}} (\widehat{\mu}_{h,i}(s') - \mu_{h,i}^0(s')) [\widehat{V}_{j,h+1}]_\alpha(s') ds' \right|, \end{aligned}$$

1026 where the first inequality is due to (6), (8) as well as Lemma 7, and the last inequality is based on
 1027 $\phi_i(s, a) \geq 0$ for any $(s, a) \in \mathcal{S} \times \mathcal{A}$ from Assumption 1. Moreover,

$$\begin{aligned}
 & \left| \int_{\mathcal{S}} \mu_{h,i}^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' - \int_{\mathcal{S}} \widehat{\mu}_{h,i}(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' \right| \\
 &= \left| \int_{\mathcal{S}} \mu_{h,i}^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' - \mathbb{1}_i^{\top} \Lambda_h^{-1} \left(\sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) [\widehat{V}_{j,h+1}]_{\alpha}(s_{h+1}^{\tau}) \right) \right| \\
 &= \left| \mathbb{1}_i^{\top} \Lambda_h^{-1} \left(\Lambda_h \int_{\mathcal{S}} \mu_h^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' - \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) [\widehat{V}_{j,h+1}]_{\alpha}(s_{h+1}^{\tau}) \right) \right| \\
 &= \left| \mathbb{1}_i^{\top} \Lambda_h^{-1} \left[\lambda_0 \int_{\mathcal{S}} \mu_h^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' + \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \left(\int_{\mathcal{S}} \widetilde{P}_h^0(s' | s_h^{\tau}, a_h^{\tau}) [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' - [\widehat{V}_{j,h+1}]_{\alpha}(s_{h+1}^{\tau}) \right) \right] \right| \\
 &= \left| \mathbb{1}_i^{\top} \Lambda_h^{-1} \left[\lambda_0 \int_{\mathcal{S}} \mu_h^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' + \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \left(\int_{\mathcal{S}} P_h^0(s' | s_h^{\tau}, a_h^{\tau}) [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' - [\widehat{V}_{j,h+1}]_{\alpha}(s_{h+1}^{\tau}) \right) \right] \right| \\
 &\quad + \left| \mathbb{1}_i^{\top} \Lambda_h^{-1} \left[\sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \left(\int_{\mathcal{S}} (\widetilde{P}_h^0(s' | s_h^{\tau}, a_h^{\tau}) - P_h^0(s' | s_h^{\tau}, a_h^{\tau})) [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' \right) \right] \right| \\
 &= \left| \mathbb{1}_i^{\top} \Lambda_h^{-1} \left[\lambda_0 \int_{\mathcal{S}} \mu_h^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' + \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \epsilon_h^{\tau}(\alpha, \widehat{V}_{j,h+1}) \right] \right| + \left| \psi H \mathbb{1}_i^{\top} \Lambda_h^{-1} \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \right|
 \end{aligned}$$

where the first equality is from (17), the third one is due to (9) and we let

$$\epsilon_h^{\tau}(\alpha, V_j) = \int_{\mathcal{S}} P_h^0(s' | s_h^{\tau}, a_h^{\tau}) [V_j]_{\alpha}(s') ds' - [V_j]_{\alpha}(s_{h+1}^{\tau}),$$

1028 for any $V_j : \mathcal{S} \rightarrow [0, H]$ and $\alpha \in [\min_s V_j(s), \max_s V_j(s)]$. Then, we have

$$\begin{aligned}
 & \left| \phi_i(s, a) \cdot (\widehat{\mu}_{h,i} - \mu_{h,i}^0) [\widehat{V}_{j,h+1}]_{\alpha} \right| \\
 &\leq \left| \phi_i(s, a) \mathbb{1}_i^{\top} \Lambda_h^{-1} \left(\lambda_0 \int_{\mathcal{S}} \mu_h^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' + \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \epsilon_h^{\tau}(\alpha, \widehat{V}_{j,h+1}) + \psi H \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \right) \right| \\
 &\leq \|\phi_i(s, a)\|_{\Lambda_h^{-1}} \underbrace{\left(\lambda_0 \left\| \int_{\mathcal{S}} \mu_h^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' \right\|_{\Lambda_h^{-1}} \right)}_{(i)} + \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \epsilon_h^{\tau}(\alpha, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}} \\
 &\quad + \underbrace{\left\| \psi H \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^{\tau}, a_h^{\tau}) \right\|_{\Lambda_h^{-1}}}_{(ii)}, \tag{35}
 \end{aligned}$$

1029 where the last inequality holds due to the Cauchy-Schwarz inequality. Moreover, the term (i) in
 1030 (35) can be further simplified to

$$(i) \leq \lambda_0 \|\Lambda_h^{-1}\|^{\frac{1}{2}} \left\| \int_{\mathcal{S}} \mu_h^0(s') [\widehat{V}_{j,h+1}]_{\alpha}(s') ds' \right\| \leq \sqrt{\lambda_0} H,$$

1031 since $|\widehat{V}_{h+1}(s)| \leq H$ for any $s \in \mathcal{S}$ and $\|\Lambda_h^{-1}\| \leq 1/\lambda_0$. And the term (ii) in (35) can be further
 1032 simplified to

$$\begin{aligned}
 (\text{ii}) &= \psi H \sqrt{\sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau)^\top \Lambda_h^{-1} \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau)} \\
 &= \psi H \sqrt{\text{Tr}(\Lambda_h^{-1} \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau)^\top)} \\
 &= \psi H \sqrt{\text{Tr}(\Lambda_h^{-1} (\Lambda_h - \lambda_0 I_d))} \\
 &\leq \psi H \sqrt{\text{Tr}(\Lambda_h^{-1} \Lambda_h)} \\
 &= \sqrt{d} \psi H.
 \end{aligned}$$

1033 Then we have

$$\begin{aligned}
 &\left| \sum_{i \in [d]} \phi_i(s, a) (\widehat{\nu}_{h,i}^{\rho, \widehat{V}_j} - \nu_{h,i}^{\rho, \widehat{V}_j}) \right| \\
 &\leq \left(\sqrt{\lambda_0} H + \max_{\alpha \in [\min_s \widehat{V}_{j,h+1}(s), \max_s \widehat{V}_{j,h+1}(s)]} \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) \right\|_{\Lambda_h^{-1}} + \sqrt{d} \psi H \right) \sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}},
 \end{aligned}$$

1034 for any $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$.

1035 B.3.5 Proof of (32)

1036 Since $\epsilon_h^\tau(\alpha, V_j)$ is 2-Lipschitz with respect to α for any $V_j : \mathcal{S} \rightarrow [0, H]$, i.e.

$$|\epsilon_h^\tau(\alpha, V_j) - \epsilon_h^\tau(\alpha^\dagger, V_j)| \leq 2|\alpha - \alpha^\dagger| \leq 2\epsilon_0.$$

1037 Therefore, for any $\alpha \in [0, H]$, one has

$$\begin{aligned}
 &\left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) (\epsilon_h^\tau(\alpha, V_j) - \epsilon_h^\tau(\alpha^\dagger, V_j)) \right\|_{\Lambda_h^{-1}}^2 \\
 &= \sum_{\tau, \tau' \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau)^\top \Lambda_h^{-1} \phi(s_h^{\tau'}, a_h^{\tau'}) \left[(\epsilon_h^\tau(\alpha, V_j) - \epsilon_h^\tau(\alpha^\dagger, V_j)) (\epsilon_h^{\tau'}(\alpha, V_j) - \epsilon_h^{\tau'}(\alpha^\dagger, V_j)) \right] \\
 &\leq \sum_{\tau, \tau' \in \mathcal{D}_h^0} \phi(s_h, a_h^\tau)^\top \Lambda_h^{-1} \phi(s_h^{\tau'}, a_h^{\tau'}) \cdot 4\epsilon_0^2 \\
 &\leq 4\epsilon_0^2 N_h^2 / \lambda_0,
 \end{aligned}$$

1038 where the last inequality is based on $\|\phi(s, a)\| \leq 1$ and $\lambda_{\min}(\Lambda_h) \geq \lambda_0$ for any $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times$
 1039 $[H]$ such that

$$\sum_{\tau, \tau' \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau)^\top \Lambda_h^{-1} \phi(s_h^{\tau'}, a_h^{\tau'}) = \sum_{\tau, \tau' \in \mathcal{D}_h^0} \|\phi(s_h^\tau, a_h^\tau)\|_2 \cdot \|\phi(s_h^{\tau'}, a_h^{\tau'})\|_2 \cdot \|\Lambda_h^{-1}\| \leq N_h^2 / \lambda_0.$$

1040 Thus,

$$\max_{\alpha \in [0, H]} \left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \left(\epsilon_h^\tau(\alpha, \widehat{V}_{j,h+1}) - \epsilon_h^\tau(\alpha^\dagger, \widehat{V}_{j,h+1}) \right) \right\|_{\Lambda_h^{-1}}^2 \leq 4\epsilon_0^2 N_h^2 / \lambda_0 \leq 4\epsilon_0^2 K^2 / \lambda_0,$$

1041 due to the fact $N_h \leq K$ for any $h \in [H]$, which completes the proof of (32).

1042 B.3.6 Proof of Lemma 11

1043 For any fixed $h \in [H]$ and $\tau \in \mathcal{D}_h^0$, we define the σ -algebra

$$\mathcal{F}_{h,\tau} = \sigma(\{(s_h^k, a_h^k)\}_{k=1}^{(\tau+1) \wedge |N_h|}, \{j_h^k, s_{h+1}^k\}_{k=1}^\tau), \quad j \in \{r, g\}.$$

As shown in Jin et al. [33, Lemma B.2], for any $\tau \in \mathcal{D}_h^0$, we have $\phi(s_h^\tau, a_h^\tau)$ is $\mathcal{F}_{h,\tau-1}$ -measurable and $\epsilon_h^\tau(\alpha, V_j)$ is $\mathcal{F}_{h,\tau-1}$ -measurable. Hence $\{\epsilon_h^\tau(\alpha, V_j)\}_{\tau \in \mathcal{D}_h^0}$ is stochastic process adapted to the filtration $\{\mathcal{F}_{h,\tau}\}_{\tau \in \mathcal{D}_h^0}$. Then, we have

$$\begin{aligned}\mathbb{E}_{\mathcal{D}_h^0}[\epsilon_h^\tau(\alpha, V_j)|\mathcal{F}] &= \int_{\mathcal{S}} P_h^0(s'|s_h^\tau, a_h^\tau)[V_j]_\alpha - \mathbb{E}_{\mathcal{D}_h^0} \left[[V_j(s_{h+1}^\tau)]_\alpha | \{(s_h^k, a_h^k)\}_{k=1}^{(\tau) \wedge N_h}, \{j_h^k, s_{h+1}^k\}_{k=1}^{\tau-1} \right] \\ &= \int_{\mathcal{S}} P_h^0(s'|s_h^\tau, a_h^\tau)[V_j]_\alpha - \mathbb{E}_{\mathcal{D}_h^0} [[V_j(s_{h+1}^\tau)]_\alpha] = 0.\end{aligned}$$

Note that $\epsilon_h^\tau(\alpha, V_j) = \int_{\mathcal{S}} P_h^0(s'|s_h^\tau, a_h^\tau)[V_j]_\alpha - [V_j(s_{h+1}^\tau)]_\alpha$ for any $V_j \in [0, H]^{\mathcal{S}}$ and $\alpha \in [0, H]$. Then, we have

$$|\epsilon_h^\tau(\alpha, V_j)| \leq H.$$

Hence, for the fixed $h \in [H]$ and all $\tau \in [H]$, the random variable $\epsilon_h^\tau(\alpha, V_j)$ is mean-zero and H -sub-Gaussian conditioning on $\mathcal{F}_{h,\tau-1}$. Then, we invoke the Lemma 3 with $\eta_\tau = \epsilon_h^\tau(\alpha, V_j)$ and $x_\tau = \phi(s_h^\tau, a_h^\tau)$. For any $\delta > 0$, we have

$$P_{\mathcal{D}} \left(\left\| \sum_{\tau \in \mathcal{D}_h^0} \phi(s_h^\tau, a_h^\tau) \epsilon_h^\tau(\alpha, V_j) \right\|_{\Lambda_h^{-1}}^2 > 2H^2 \log \left(\frac{\det(\Lambda_h^{1/2})}{\delta \det(\lambda_0 I_d)^{1/2}} \right) \right) \leq \delta.$$

Together with the facts that $\det(\Lambda_h^{1/2}) = (\lambda_0 + N_h)^{d/2}$ and $\det(\lambda_0 I_d)^{1/2} = \lambda_0^{d/2}$, we can conclude the proof of Lemma 11.

B.4 Proof of Corollary 1

Before continuing, we introduce some additional notations that will be used in the following analysis. For any $(h, i) \in [H] \times [d]$, define $\Phi_{h,i}^* : \mathcal{S} \rightarrow \mathbb{R}^{d \times d}$ and $b_{h,i}^* : \mathcal{S} \rightarrow \mathbb{R}$ by

$$\begin{aligned}\Phi_{h,i}^*(s) &= (\phi_i(s, \pi_h^*(s)) \mathbb{1}_i)(\phi_i(s, \pi_h^*(s)) \mathbb{1}_i)^\top \in \mathbb{R}^{d \times d}, \\ b_{h,i}^*(s) &= (\phi_i(s, \pi_h^*(s)) \mathbb{1}_i)^\top \Lambda_h^{-1} (\phi_i(s, \pi_h^*(s)) \mathbb{1}_i).\end{aligned}$$

With these notations in hand and recalling (11) in Theorem 1, one has

$$\begin{aligned}V_{r,1}^{*,\rho}(\zeta) - V_{r,1}^{\hat{\pi},\rho}(\zeta) &\leq 2\gamma_0 \sum_{h=1}^H \sum_{i=1}^d \sup_{d_h^* \in \mathcal{D}_h^*} \mathbb{E}_{s \sim d_h^*} \sqrt{b_{h,i}^*(s)} \\ &\leq 2\gamma_0 \sum_{h=1}^H \sum_{i=1}^d \sup_{d_h^* \in \mathcal{D}_h^*} \sqrt{\mathbb{E}_{s \sim d_h^*} b_{h,i}^*(s)} \\ &= 2\gamma_0 \sum_{h=1}^H \sup_{d_h^* \in \mathcal{D}_h^*} \sum_{i=1}^d \sqrt{\mathbb{E}_{s \sim d_h^*} b_{h,i}^*(s)},\end{aligned}\tag{36}$$

where the second inequality is due to the Jensen's inequality and concavity.

In the following, we will control the key term $\sum_{i=1}^d \sqrt{\mathbb{E}_{s \sim d_h^*} b_{h,i}^*(s)}$ for any $d_h^* \in \mathcal{D}_h^*$. Before continuing, we first denote

$$\mathcal{C}_h^b = \{(s, a) : d_h^b(s, a) > 0\}.$$

Considering any (s, a) s.t. $d_h^b(s, a) > 0$ and from Lemma 8, the following lower bound holds with probability at least $1 - 3\delta$, i.e.,

$$N_h(s, a) \geq \frac{K d_h^b(s, a)}{8} - 5 \sqrt{K d_h^b(s, a) \log \left(\frac{KH}{\delta} \right)} \geq \frac{K d_h^b(s, a)}{16},\tag{37}$$

as long as

$$K \geq c_0 \frac{\log(KH/\delta)}{d_{\min}^b} \geq c_0 \frac{\log(KH/\delta)}{d_h^b(s, a)}\tag{38}$$

1064 for some sufficiently large c_0 and $d_{\min}^b = \min_{h,s,a} \{d_h^b(s,a) : d_h^b(s,a) > 0\}$. Therefore,

$$\begin{aligned}\Lambda_h &= \sum_{(s,a) \in \mathcal{C}_h^b} N_h(s,a) \phi(s,a) \phi(s,a)^\top + I_d \\ &\succeq \sum_{(s,a) \in \mathcal{C}_h^b} \frac{K d_h^b(s,a)}{16} \phi(s,a) \phi(s,a)^\top + I_d \\ &\succeq \frac{K}{16} \mathbb{E}_{d_h^b} [\phi(s,a) \phi(s,a)^\top] + I_d.\end{aligned}$$

1065 From Assumption 3,

$$\mathbb{E}_{d_h^b} [\phi(s,a) \phi(s,a)^\top] \succeq \max_{P \in \mathcal{P}^\rho(P^0)} \frac{d \cdot \min\{\mathbb{E}_{d_h^{*,P}} \phi_i^2(s,a), 1/d\}}{C_{\text{rob}}^*} \mathbb{1}_{i,i}, \quad \forall i \in [d]$$

1066 Thus, for any $i \in [d]$,

$$\Lambda_h \succeq I_d + \frac{Kd \cdot \min\{\mathbb{E}_{d_h^*} \phi_i^2(s, \pi_h^*(s)), 1/d\}}{16C_{\text{rob}}^*} \cdot \mathbb{1}_{i,i}.$$

1067 Here, $\mathbb{1}_{i,j}$ represents a matrix with the (i,j) -th coordinate as 1 and all other elements as 0. Conse-
1068 quently,

$$\begin{aligned}\mathbb{E}_{s \sim d_h^*} b_{h,i}^*(s) &= \mathbb{E}_{s \sim d_h^*} \text{Tr}(\Phi_{h,i}^*(s) \Lambda_h^{-1}) = \text{Tr}(\mathbb{E}_{s \sim d_h^*} \Phi_{h,i}^*(s) \Lambda_h^{-1}) \\ &\leq \frac{\mathbb{E}_{d_h^*} \phi_i^2(s, \pi_h^*(s))}{1 + Kd \cdot \min\{\mathbb{E}_{d_h^*} \phi_i^2(s, \pi_h^*(s)), 1/d\} / 16C_{\text{rob}}^*},\end{aligned}\quad (39)$$

1069 where the second equality is because the trace is a linear mapping and the last inequality holds by
1070 Lemma 5. We further define $\mathcal{E}_{h,\text{larger}} = \{i : \mathbb{E}_{(s,a) \sim d_h^*} \phi_i^2(s,a) \geq \frac{1}{d}\}$. Due to Assumption 1, we first
1071 claim that

$$|\mathcal{E}_{h,\text{larger}}| \leq \sqrt{d}, \quad (40)$$

1072 where the proof can be found at the end of this subsection.

1073 By utilizing Assumption 3, we discuss the following three cases.

- 1074 • If $\mathbb{E}_{(s,a) \sim d_h^*} \phi_i^2(s,a) = 0$ ($i \notin \mathcal{E}_{h,\text{larger}}$), it is easily observed that (39) can be controlled by
1075 $\langle d_h^*, b_{h,i}^* \rangle \leq 0$.
- 1076 • If $0 < \mathbb{E}_{(s,a) \sim d_h^*} \phi_i^2(s,a) \leq \frac{1}{d}$ ($i \notin \mathcal{E}_{h,\text{larger}}$), we have

$$(39) \leq \frac{16C_{\text{rob}}^* \cdot \mathbb{E}_{d_h^*} \phi_i^2(s, \pi_h^*(s))}{Kd \cdot \mathbb{E}_{d_h^*} \phi_i^2(s, \pi_h^*(s))} = \frac{16C_{\text{rob}}^*}{Kd}.$$

- 1077 • If $i \in \mathcal{E}_{h,\text{larger}}$, i.e., $\frac{1}{d} \leq \mathbb{E}_{(s,a) \sim d_h^*} \phi_i^2(s,a) \leq 1$, we have

$$(39) \leq \frac{16C_{\text{rob}}^* \cdot \mathbb{E}_{d_h^*} \phi_i^2(s, \pi_h^*(s))}{K} \leq \frac{16C_{\text{rob}}^*}{K},$$

1078 where the last inequality holds due to $\phi_i^2(s, \pi_h^*(s)) \leq 1$.

1079 Summing up the above three cases and (40), we have

$$\begin{aligned}\sum_{i=1}^d \sqrt{\mathbb{E}_{s \sim d_h^*} b_{h,i}^*(s)} &\leq \sum_{i \in \mathcal{E}_{h,\text{larger}}} \sqrt{\mathbb{E}_{s \sim d_h^*} b_{h,i}^*(s)} + \sum_{i \notin \mathcal{E}_{h,\text{larger}}} \sqrt{\mathbb{E}_{s \sim d_h^*} b_{h,i}^*(s)} \\ &\leq |\mathcal{E}_{h,\text{larger}}| \sqrt{\frac{16C_{\text{rob}}^*}{K}} + |d - \mathcal{E}_{h,\text{larger}}| \sqrt{\frac{16C_{\text{rob}}^*}{Kd}} \\ &\leq 8\sqrt{C_{\text{rob}}^*} \sqrt{\frac{d}{K}}.\end{aligned}$$

1080 Together with (36) and setting $\gamma_0 = 6\sqrt{d}H\sqrt{\log(3HK/\delta)}$, one obtains

$$\begin{aligned} V_{r,1}^{\star,\rho}(\zeta) - V_{r,1}^{\hat{\pi},\rho}(\zeta) &\leq 2\gamma_0 \sum_{h=1}^H \sup_{d_h^{\star} \in \mathcal{D}_h^{\star}} \sum_{i=1}^d \sqrt{\mathbb{E}_{s \sim d_h^{\star}} b_{h,i}^{\star}(s)} \\ &\leq 96dH^2 \sqrt{C_{\text{rob}}^{\star}/K} \sqrt{\log(3HK/\delta)}, \end{aligned}$$

1081 with probability at least $1 - 4\delta$, as long as $K \geq c_0 \frac{\log(KH/\delta)}{d_{\min}^b}$ for some universal constant c_0 .

1082 **Proof of (40).** Let $\tilde{\mathcal{E}}_{h,\text{larger}} = \{i : \mathbb{E}_{(s,a) \sim d_h^{\star}} \phi_i(s, a) \geq \frac{1}{\sqrt{d}}\}$.

1083 • We first show that $|\tilde{\mathcal{E}}_{h,\text{larger}}|$ should be no larger than $\sqrt{d}$ by contradiction. Suppose
1084 $|\tilde{\mathcal{E}}_{h,\text{larger}}| > \sqrt{d}$. Then, there are more than $\sqrt{d}$ coordinates of $\mathbb{E}_{(s,a) \sim d_h^{\star}} \phi(s, a) \in \mathbb{R}^d$
1085 that is larger than $1/\sqrt{d}$. In other words,

$$\sum_{i \in \tilde{\mathcal{E}}_{h,\text{larger}}} \mathbb{E}_{(s,a) \sim d_h^{\star}} \phi_i(s, a) > 1,$$

1086 which is equivalent to

$$\max_{(s,a) \in \mathcal{S} \times \mathcal{A}} \|\phi(s, a)\|_1 \geq \mathbb{E}_{(s,a) \sim d_h^{\star}} \|\phi(s, a)\|_1 \geq \mathbb{E}_{(s,a) \sim d_h^{\star}} \sum_{i \in \tilde{\mathcal{E}}_{h,\text{larger}}} \phi_i(s, a) > 1,$$

1087 where the last inequality is from the linearity of the expectation mapping. It contradicts to
1088 our Assumption 2, which implies $\|\phi(s, a)\|_1 = 1$ for any $(s, a) \in \mathcal{S} \times \mathcal{A} \times [H]$.

1089 • Then, we show that $\tilde{\mathcal{E}}_{h,\text{larger}} \subseteq \mathcal{E}_{h,\text{larger}}$: For every element $i \in \tilde{\mathcal{E}}_{h,\text{larger}}$, we have

$$\frac{1}{d} \leq (\mathbb{E}_{(s,a) \sim d_h^{\star}} \phi_i(s, a))^2 \leq \mathbb{E}_{(s,a) \sim d_h^{\star}} \phi_i^2(s, a),$$

1090 where the second inequality is due to the Jensen's inequality. Thus, $\tilde{\mathcal{E}}_{h,\text{larger}} \subseteq \mathcal{E}_{h,\text{larger}}$.

1091 Combining these two arguments, we show that $|\mathcal{E}_{h,\text{larger}}| \geq \sqrt{d}$.

1092 B.5 Proof of Corollary 2

1093 We first establish the following lemma to control the sub-optimality, under the full feature coverage.

1094 **Lemma 12.** Consider $\delta \in (0, 1)$. Suppose Assumption 2, Assumption 4 and all conditions in Lemma
1095 6 hold. For any $h \in [H]$, if $N_h \geq \max\{512 \log(2Hd/\delta)/\kappa^2, 4/\kappa\}$, we have

$$\sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}} \leq \frac{2}{\sqrt{N_h \kappa}}, \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A},$$

1096 with probability exceeding $1 - \delta$.

1097 *Proof.* From Lemma 6 and Assumption 4, one has

$$\|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}} \leq \frac{2\phi_i(s, a)}{\sqrt{N_h \kappa}}, \quad \forall (i, s, a) \in [d] \times \mathcal{S} \times \mathcal{A},$$

1098 as long as $N_h \geq \max\{512 \log(2Hd/\delta)/\kappa^2, 4/\kappa\}$. In addition,

$$1 = \int_{\mathcal{S}} P_h^0(s'|s, a) ds' = \int_{\mathcal{S}} \phi(s, a)^{\top} \mu_h^0(s') ds' = \sum_{i=1}^d \phi_i(s, a) \int_{\mathcal{S}} \mu_{h,i}^0(s') ds' = \sum_{i=1}^d \phi_i(s, a),$$

1099 where the last equality is implied by Assumption 2. Therefore,

$$\sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}} \leq \sum_{i=1}^d \frac{2\phi_i(s, a)}{\sqrt{N_h \kappa}} \leq \frac{2}{\sqrt{N_h \kappa}}.$$

1100 □

1101 From (37), we have $N_h \geq \frac{K}{16}$ with probability exceeding $1 - 3\delta$, as long as K obeys (38). Together
 1102 will Lemma 12, with probability exceeding $1 - 4\delta$, one has

$$\sum_{i=1}^d \|\phi_i(s, a) \mathbb{1}_i\|_{\Lambda_h^{-1}} \leq \frac{8}{\sqrt{K\kappa}}, \forall (s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H],$$

1103 as long as $K \geq \max\{c_0 \log(2Hd/\delta)/\kappa^2, c_0 \log(KH/\delta)/d_{\min}^b\}$ for some sufficiently large universal
 1104 constant c_0 . It follows Theorem 1 that

$$\text{Regret}(\hat{\pi}; \zeta, \mathcal{P}^\rho) \leq 96\sqrt{d}H^2 \sqrt{\frac{\log(3HK/\delta)}{K\kappa}},$$

1105 which completes the proof.

1106

□